

INTERVIEW PREPARATION GUIDE

Java, Spring Boot & DSA Infosys

Core Java · Spring Boot · JPA · DSA · SQL · HR

Prepared for Sujith Mondithoka · Interview: Saturday 26 September

How to use this guide. Work the sections in order. The plan on the next page is timed — if you fall behind, drop from the bottom, never from the top.

● **Red dot** = high priority, almost certainly asked. ●● = your differentiator. ● **Amber** = worth knowing.

✓ = self-check before moving on. ⚠ = common trap or something to verify.

Contents

Plan	Infosys — Java, Spring Boot and DSA Prep	3
01	Core Java and Collections	8
02	Java 8 and Functional Programming	19
03	Spring Boot	27
04	JPA, Hibernate and the N+1 Problem	38
05	Microservices	47
06	Kafka and RabbitMQ	54
07	Spring Security and JWT	59
08	SQL and Query Optimisation	64
09	DSA Part 1 — Patterns, Arrays, Strings, Hashing	70
10	DSA Part 2 — Linked Lists, Stacks, Recursion, Searching, Trees	79
11	Backend System Design	86
12	Talking About Your Experience	90
13	HR, Managerial and the Offer Conversation	95
14	Delivery, Code Review and Working With Clients	100
15	JUnit, Mockito, Build Tools and Agile	107
16	Final Cheat Sheet — SATURDAY MORNING ONLY	115

Infosys — Java, Spring Boot and DSA Prep

For Sujith Mondithoka · Interview: Saturday 26 September · 4 days from today

How this guide is written

Every topic follows the same four steps:

1. **The context** — what it is and what problem it solves, in plain words.
2. **A simple example** — small code you can follow.
3. **Why it matters** — where it is used in real work.
4. **"Say this"** — the answer to give in the room.

Every question has a written answer in the words you would say, and most carry a **real-world example** anchored to the enterprise banking platform you actually worked on.

Do not memorise them word for word. Learn the shape, so that when the question comes out slightly differently you can still answer it.

1. The job description, decoded

Service line: **Engineering Services**. Preferred skills: **Core Java** and **Spring Boot**.

This is a build role, and the spec is unusually specific — which is good news, because it tells you exactly what to revise.

What it names, and where it is covered

The JD asks for	Your position	File
Core Java (8/11/17+)	⚠ You list Java 8 — know what 11 and 17 added	01 , 02
Spring Boot	✓ Strong — you built two backends	03
Spring MVC	Know the DispatcherServlet flow	03 , Q3b
Spring Data JPA, Hibernate	✓ Strong	04
Spring Security	✓ You did RBAC on the card flow	07
Microservices architecture	⚠ "Strong understanding" — read Q7, answer honestly	05
RESTful APIs	✓ Strong — consumed by a 14-screen React frontend	03 , 11
SQL: Oracle, MySQL, PostgreSQL, SQL Server	✓ PostgreSQL and MySQL. The skills transfer; say so	08

The JD asks for	Your position	File
Git	✓ Git and Bitbucket	15 , Q8
Maven or Gradle	✓ Maven. Know the lifecycle	15 , Q7
JUnit, Mockito	⚠ Your resume says JUnit, not Mockito. Close this gap	15 , Part A
Cloud platforms (integration)	⚠ Docker yes, cloud no. Answer honestly	12 , 15 Q13
Code reviews, coding standards	✓ You did these	14
Agile ceremonies	✓ Scrum and Jira — have a line on each ceremony	15 , Part C
Technical documentation	✓ Swagger. And Liquibase for schema	15 , Q11
Production deployment support	✓ Bamboo CI/CD, SIT/UAT	15 , Q12

The four gaps, and the pattern for all of them

Mockito · cloud platforms · Java 11/17 · microservices depth.

None is disqualifying. The answer pattern is the same every time: **name the boundary, then show the understanding.**

"My mocking has been lighter than my JUnit usage, so I would not overstate it — but I understand the model: mock the collaborators you do not own, stub with `when/thenReturn`, verify interactions." *...then demonstrate it.*

At one to two years that is a strong answer. Claiming depth and then failing the second question is not — and the second question always comes.

Mockito is the one to actually fix this week, not just frame. It is named in the JD, it takes an hour to learn properly, and it is the cheapest marks available. File 15, Part A.

Where you are genuinely strong

- **You built the backends**, not just consumed them — Spring Boot, Spring Data JPA and PostgreSQL for two applications on a platform used by 10,000+ corporate clients.
- **Real numbers**: card processing time down **60%**, an audit system writing **50,000+ transactions a day**, 1,000+ monthly reports.
- **Unusual depth for your level** — Camunda BPMN, Spring AOP with `@Aspect`, Liquibase, JasperReports, FileNet, Kafka and RabbitMQ. Most candidates at two years have CRUD APIs.
- **The JD's whole responsibilities list is things you have done**: requirements analysis with BAs, code reviews, troubleshooting and optimisation, Agile ceremonies, documentation, supporting releases.

And DSA, which you asked about

Infosys almost always includes a coding question, and it is what candidates most often walk into cold. Set expectations correctly: **arrays, strings, HashMap counting, two pointers, basic recursion, sorting and searching**, sometimes a linked list or stack. Easy to lower-medium. **They essentially never ask dynamic programming or graphs** — if you are on DP on Friday night, stop.

What they check: compiling Java without an IDE, the right collection and why, edge cases, **stating complexity**, and **thinking out loud**. The last two are free marks most candidates drop. Files 09 and 10, and every solution in them is compile-tested.

2. What the interview will probably look like

Round 1 — Technical (45-60 min). Core Java, collections, Java 8, Spring Boot, JPA, SQL, plus a coding question. → Files 01-04, 08, 09, 10.

Round 2 — Technical / project (30-45 min). A deep dive into what you built, plus microservices, messaging and design. → Files 05, 06, 11, 12.

Round 3 — Managerial and HR (20-30 min). Delivery process, behaviour, the gap, salary, location. → Files 13, 14.

⚠ This is my prediction from how these interviews usually run, not something I can confirm. Ask your recruiter what the rounds are — it is a normal question and it costs nothing.

3. The four-day plan

Tuesday (today) — Java fundamentals

The thing they screen hardest on.

Block	Topic	File
0:15	Read this file	README.md
1:45	Core Java and Collections ●● — HashMap above all	01-core-java.md
1:00	Java 8 — streams, lambdas, Optional ●	02-java8-functional.md
0:45	DSA: complexity table + the hashing pattern ●● — type them	09-dsa-arrays-strings.md

Wednesday — Spring Boot and the data layer

Block	Topic	File
1:30	Spring Boot ●● — DI, annotations, bean scope, @ControllerAdvice	03-spring-boot.md
1:15	JPA, Hibernate, N+1, @Transactional ●●	04-jpa-hibernate.md

Block	Topic	File
1:00	SQL and indexing ●	08-sql-and-databases.md
0:45	DSA: two pointers + sliding window ● — type them	09-dsa-arrays-strings.md
0:30	Spring MVC — the DispatcherServlet flow ●	03-spring-boot.md Q3b

Thursday — DSA day, and your stories

Block	Topic	File
1:30	DSA Part 1, finish and re-do the ones you got wrong ●●	09-dsa-arrays-strings.md
1:30	DSA Part 2 — linked lists, stacks, recursion, binary search ●	10-dsa-structures.md
1:00	Your experience ●● — the AOP story, the gap, "why backend"	12-your-experience.md
0:45	Microservices ● — especially Q7, the honest answer	05-microservices.md

Friday — the rest, and rehearsal

Block	Topic	File
1:15	JUnit, Mockito, Maven, Agile ●● — the JD names all of it	15-testing-build-agile.md
0:45	Spring Security and JWT ● — named in the JD	07-spring-security-jwt.md
0:45	Delivery and code review ●	14-delivery-and-code-review.md
0:30	HR and managerial ● — say these out loud	13-hr-and-behavioural.md
0:45	Re-do your five weakest DSA problems from memory ●●	09 , 10
0:30	Skim: messaging, then system design	06 , 11
—	Sleep properly. Do not cram Friday night.	—

Saturday morning

45 minutes on [16-final-cheatsheet.md](#) and nothing else. Then say your intro, the gap answer and the AOP story out loud, twice. **Learn nothing new.**

If you fall behind

Drop in this order: system design → messaging → Spring Security → microservices beyond Q7.

Never drop: Core Java, Spring Boot, JPA, DSA Part 1, file 12, or **file 15 Part A (Mockito)** — that last one is named in the JD and missing from your resume.

4. The files

File	Topic	Priority
01-core-java.md	OOP, Collections, HashMap internals, String, exceptions	● Highest
02-java8-functional.md	Streams, lambdas, functional interfaces, Optional	● High
03-spring-boot.md	IoC, DI, annotations, bean scopes, auto-configuration	● Highest
04-jpa-hibernate.md	JPA, lazy vs eager, N+1 , transactions, caching	● Highest
05-microservices.md	Patterns, resilience, saga — read Q7	● Medium
06-kafka-rabbitmq.md	Messaging, queue vs event log	● Medium
07-spring-security-jwt.md	Authentication, RBAC, JWT — named in the JD	● High
08-sql-and-databases.md	Joins, indexing, EXPLAIN, optimisation	● High
09-dsa-arrays-strings.md	Patterns, complexity, arrays, strings, hashing	● Highest
10-dsa-structures.md	Linked lists, stacks, recursion, search, trees	● High
11-system-design.md	Design framework and worked examples	● Medium
12-your-experience.md	Your stories, the gap, "why backend"	● Highest
13-hr-and-behavioural.md	HR round, salary, questions to ask	● High
14-delivery-and-code-review.md	Code review, estimation, working with clients	● High
15-testing-build-agile.md	JUnit, Mockito, Maven, Git, Agile ceremonies	● Highest
16-final-cheatsheet.md	One page. Saturday morning only.	● Read last

5. Three rules for the room

1. In the coding round, keep talking. Silence reads as stuck. Narrate: *"I'll count with a HashMap so the lookup is $O(1)$, then find the first with a count of one."*

2. Always say the brute force first, with its cost. *"Two nested loops is $O(n^2)$ — I can do better with a map."* Then improve it. Jumping straight to a memorised optimal solution shows nothing.

3. Give the why, not just the what. "HashMap stores key-value pairs" is a textbook answer. "It gives $O(1)$ lookup because it hashes the key to a bucket instead of scanning" is an engineer's answer. Same length, completely different impression.

01 · Core Java and Collections

Time needed: 105 minutes

This is where Infosys screens hardest. Service company interviews go deep on fundamentals, because fundamentals are what transfer across the many different client projects they will put you on. Expect rapid-fire questions here.

If you only master one topic in this file, make it HashMap. It is the single most asked Java interview question in India.

Part A · OOP — the four pillars

You will be asked to name them and give an example of each. Have a real example ready, not a "Dog extends Animal" one.

| Q1. Encapsulation

What it is: keeping data private and exposing it only through methods.

```
public class Account {  
    private double balance;                // nobody can touch this directly  
  
    public void deposit(double amount) {  
        if (amount <= 0) throw new IllegalArgumentException("Amount must be positive");  
        this.balance += amount;           // the rule is enforced here  
    }  
  
    public double getBalance() { return balance; }  
}
```

Why it matters: if `balance` were public, any code anywhere could set it to a negative number and no validation would run. Encapsulation means the object protects its own rules. In a banking system that is not a style preference.

Say this.

"Encapsulation means the object owns its data and its rules. The fields are private and the only way in is through methods that validate. In the card system, a credit limit could not just be assigned — it had to go through a method that checked the approval status and the band the customer qualified for. If that field were public, any code anywhere could set it and skip every check, and you would never find out where it happened."

Q2. Inheritance

One class reuses another's behaviour with `extends`. A `SavingsAccount extends Account` gets `deposit()` for free and adds interest calculation.

The follow-up: "Why does Java not support multiple inheritance of classes?"

"Because of the diamond problem. If B and C both extend A and override the same method, and D extends both, the compiler cannot decide which version D inherits. Java avoids that by allowing only one superclass. You can implement multiple interfaces, and since Java 8 interfaces can have default methods — so if two interfaces give the same default method, the compiler forces you to override it and resolve the conflict explicitly."

Q3. Polymorphism •

"One interface, many implementations." Two kinds, and interviewers want both.

	Overloading (compile-time)	Overriding (runtime)
What changes	Same method name, different parameters	Same signature, different class
Decided	At compile time, by the compiler	At runtime, by the actual object
Also called	Static polymorphism	Dynamic polymorphism

```
// Overloading - same name, different parameter lists
void notify(String email) { }
void notify(String email, String sms) { }

// Overriding - the subclass replaces the parent's behaviour
class Notifier { void send() { System.out.println("generic"); } }
class EmailNotifier extends Notifier { @Override void send() { System.out.println("email"); } }

Notifier n = new EmailNotifier();
n.send(); // "email" - decided by the OBJECT, not the reference type
```

That last line is the whole point of runtime polymorphism, and it is what makes Spring's dependency injection work: the field is declared as an interface, and the concrete implementation is chosen at runtime.

Real-time example. In the notification flow, the service holds a `List<Notifier>` containing an email notifier and an SMS notifier. It calls `notifier.send(...)` on each without knowing or caring which is which. When a push notifier is added later, the sending code does not change at all — that is polymorphism doing real work.

Trap: "Can you override a static method?" → No. Static methods belong to the class, not the instance, so they are **hidden**, not overridden. Which one runs is decided by the reference type, not the object.

Say this.

"Overloading is compile time — same method name, different parameters, and the compiler picks. Overriding is runtime — a subclass replaces the parent's behaviour, and which one runs depends on the actual object, not the reference type.

Overriding is what makes dependency injection work. My service holds a `Notifier` reference, and at runtime it is an email or SMS implementation. The service never changes when a new channel is added."

Q4. Abstraction

Hiding *how* something works and exposing *what* it does.

● Abstract class vs interface — near-certain question

	Abstract class	Interface
Fields	Can have normal instance fields	Only <code>public static final</code> constants
Constructor	Yes	No
Methods	Abstract and concrete	Abstract, plus <code>default</code> and <code>static</code> since Java 8
Inheritance	One only	Many
Access modifiers	Any	Methods are public

When to use which — say it this way:

"An abstract class is for an 'is-a' relationship where the subclasses share state and some common implementation — a base `Transaction` class holding amount and timestamp. An interface is a contract about capability with no shared state, like `Auditable` or `PaymentProcessor`. Since I can implement many interfaces but extend only one class, I default to interfaces and use an abstract class when there is real shared code and fields."

Real-time example — say this if they ask for one.

"In the card system, `PaymentProcessor` was an **interface** with implementations for different rails — NEFT, RTGS, IMPS. Each one processes a payment completely differently, so there is no shared code to inherit, only a shared contract. The service just depends on `PaymentProcessor` and Spring injects the right one.

`BaseTransaction` was an **abstract class**, because every transaction genuinely has an amount, a currency, a timestamp and an audit trail, plus a common `validate()` that all of them run. That is shared state and shared code, which an interface cannot give me."

Part B · Collections ●●

Q5. The map of the framework

Collection (interface)

└ List	- ordered, allows duplicates	→ ArrayList, LinkedList, Vector
└ Set	- no duplicates	→ HashSet, LinkedHashSet, TreeSet
└ Queue	- FIFO processing	→ PriorityQueue, ArrayDeque

Map (separate, NOT a Collection) - key/value → HashMap, LinkedHashMap, TreeMap, Hashtable

⚠ `Map` does **not** extend `Collection`. It stores pairs, not single elements. A small detail interviewers like to check.

Say this — how you choose a collection.

"I pick by the access pattern. If I need order and index access, `List`, and `ArrayList` unless I have a reason. If I need uniqueness, `Set`. If I need lookup by a key, `Map`. Then the second question is whether ordering matters — `LinkedHashSet` or `TreeSet` instead of `HashSet` — and whether it is accessed by multiple threads, which means `ConcurrentHashMap` rather than `HashMap`."

Real-time example. In the recall batch: a `List` for the page of transactions being processed, a `Set` of already-processed reference numbers so a retry does not double process, and a `Map` of status code to description loaded once for the report.

Q6. ArrayList vs LinkedList ●

	ArrayList	LinkedList
Structure	Growable array	Doubly linked list
Get by index	O(1)	O(n) — must walk the chain
Add/remove at end	O(1) amortised	O(1)
Add/remove in middle	O(n) — shifts elements	O(1) <i>once you are at the node</i>
Memory	Less	More — each node stores two pointers

Say this:

"I use ArrayList by default. Random access is O(1) and the elements sit together in memory, which is cache friendly. LinkedList only wins when I am inserting or removing frequently in the middle while already holding a reference to that position. In practice that is rare, and even for a queue I would use ArrayDeque."

Real-time example. A recall report loads a page of transactions and then iterates them to build the output — read-heavy, index access, no insertion in the middle. That is an `ArrayList`. I have not had a case in these systems where `LinkedList` was the right answer.

How does ArrayList grow? It starts at capacity 10, and when full it creates a new array about **1.5 times** the size and copies everything across. That copy is why `new ArrayList<>(expectedSize)` matters when you know the size up front.

| Q7. ●● How does HashMap work internally?

Learn this properly. It is the most asked Java question in Indian interviews.

Storing a value — `map.put(key, value)`

1. Java calls `key.hashCode()` to get an int.
2. HashMap applies its own **hash spreading** on top: `h ^ (h >>> 16)`. This mixes the high bits into the low bits, because the next step only uses the low bits.
3. It finds the bucket: `index = (n - 1) & hash`, where `n` is the array length. Since `n` is always a power of two, this is a fast bitwise AND instead of a modulus.
4. If the bucket is empty, it stores the entry there.
5. If the bucket already has entries — a **collision** — it walks them comparing with `equals()`. Same key means replace the value; otherwise append the new entry.

The structure

Internally it is an **array of buckets**, and each bucket holds a **linked list** of entries. Since **Java 8**, once a single bucket exceeds **8 entries** (and the table is at least 64 long) that list is converted into a **balanced red-black tree**, turning worst-case lookup from $O(n)$ into $O(\log n)$. Below 6 entries it converts back.

Resizing

Default capacity **16**, load factor **0.75**. When size passes $16 \times 0.75 = 12$, the array doubles to 32 and everything is **rehashed** into new buckets. Resizing is expensive, which is why you presize a map you know will be large.

Complexity

$O(1)$ average for get and put. $O(\log n)$ worst case since Java 8, thanks to treeification.

Real-time example — where you actually used one

"Two places. In the report generation I loaded the reference data once — branch codes and status descriptions — into a `HashMap` keyed by code, so the loop could look each one up in $O(1)$ instead of hitting the database per row. And on the Kafka consumer side, a map of processed message IDs is how you make a consumer idempotent: check the map before acting, so a redelivered message is a no-op."

● The follow-up that catches people: equals and hashCode

"If you use a custom object as a key you must override both `hashCode()` and `equals()`. The contract is that **equal objects must have equal hash codes**. If I override `equals` but not `hashCode`, two objects that are equal can produce different hash codes, land in different buckets, and the map will never find the entry I stored. That is also why `HashMap` keys should be **immutable** — if a field used in the hash changes after insertion, the entry is effectively lost, because it is sitting in a bucket that no longer matches its hash."

Q8. HashMap vs Hashtable vs ConcurrentHashMap ●

	HashMap	Hashtable	ConcurrentHashMap
Thread safe	No	Yes	Yes
How	—	Locks the whole map	Locks only the affected bucket
Null key/value	One null key, many null values	Neither	Neither
Performance	Fastest single-threaded	Slow — everything serialises	Fast under concurrency
Status	Standard	Legacy, do not use	Standard for concurrent use

"Hashtable synchronises every method, so all threads queue behind one lock. That is why it is effectively obsolete. `ConcurrentHashMap` locks at bucket level, so threads writing to different buckets do not block each other, which is why it scales."

Real-time example. A Spring `@Service` is a **singleton**, so any collection held in a field is shared by every request thread at once. If I keep an in-memory counter or cache in a service — say a per-branch request count — a plain `HashMap` will corrupt under load and can even spin forever on an older JDK. That field has to be a `ConcurrentHashMap`.

Q9. HashSet vs LinkedHashSet vs TreeSet

	Ordering	Backing structure	Complexity
HashSet	None guaranteed	HashMap	O(1)
LinkedHashSet	Insertion order	HashMap + linked list	O(1)
TreeSet	Sorted	Red-black tree	O(log n)

Useful detail: `HashSet` is literally a `HashMap` where every value is the same dummy object. That is why its rules about `hashCode` and `equals` are identical.

Real-time example. `HashSet` for "which account numbers have already been processed in this batch" — I only care about membership. `TreeSet` when the order matters, like interest rate slabs that must be walked from lowest to highest to find the applicable band. `LinkedHashSet` when I need uniqueness but the output must stay in the order the records arrived, which matters for a report a human reads.

Say this.

"All three guarantee uniqueness; the difference is ordering and cost. `HashSet` is $O(1)$ with no order guarantee. `LinkedHashSet` is also $O(1)$ but keeps insertion order, which matters when a human reads the output. `TreeSet` keeps them sorted, which costs $O(\log n)$ because it is a red-black tree. I use `HashSet` by default and only pay for ordering when I actually need it."

Q10. Fail-fast vs fail-safe iterators

"Fail-fast iterators, like `ArrayList`'s and `HashMap`'s, throw `ConcurrentModificationException` if the collection is structurally modified while iterating. They track a `modCount` and check it on each `next()`. Fail-safe iterators, like `ConcurrentHashMap`'s or `CopyOnWriteArrayList`'s, work on a snapshot or allow concurrent access, so they do not throw — but they may not reflect the very latest changes."

How do you remove safely while iterating? Use `iterator.remove()`, or `collection.removeIf(predicate)`. Never `list.remove()` inside a for-each loop.

Q11. Comparable vs Comparator ●

```
// Comparable - the class defines its OWN natural ordering. One only.
class Txn implements Comparable<Txn> {
    public int compareTo(Txn other) { return this.date.compareTo(other.date); }
}

// Comparator - ordering defined OUTSIDE the class. As many as you like.
Comparator<Txn> byAmount = Comparator.comparing(Txn::getAmount);
list.sort(byAmount.reversed());
```

"Comparable is the natural order and lives inside the class, so there can be only one. Comparator lives outside, so I can have several — sort transactions by date on one screen and by amount on another without touching the entity."

Real-time example. `Transaction` implements `Comparable` on `createdAt`, because newest-first is its natural order everywhere in the application. But the recall report

screen lets the user sort by amount, by status or by customer name — three `Comparator`s, none of which required changing the entity.

Part C · Strings, immutability and memory

Q12. Why is String immutable? ●

Once created, a String's value can never change. `s.concat("x")` returns a **new** String.

The four reasons — give at least two:

1. **String pool.** Java reuses identical literals to save memory. That is only safe if nobody can change them underneath other references.
2. **Security.** Usernames, file paths and DB URLs are Strings. If they were mutable, a value could be validated and then changed before use.
3. **Thread safety.** Immutable objects can be shared across threads with no locking.
4. **HashCode caching.** String caches its hash, which makes it a fast HashMap key. That is only valid if the value cannot change.

Real-time example. Account numbers and transaction reference numbers are Strings used as map keys and passed between services. If a String were mutable, code holding a reference could change an account number *after* it had been validated and after it had been used to place an entry in a map — the entry would then be unreachable. Immutability is what makes them safe to share.

Say this.

"Because too much depends on a String not changing under you. The string pool reuses identical literals, which is only safe if nobody can modify one. Strings are used for account numbers, file paths and connection URLs, so if they were mutable a value could be validated and then changed before it was used. And String caches its own hash code, which is what makes it a fast HashMap key — that cache is only valid because the value is fixed."

Q13. String vs StringBuilder vs StringBuffer

	Mutable?	Thread safe	Use when
String	No	Yes (immutable)	Fixed values
StringBuilder	Yes	No	Building strings — the default
StringBuffer	Yes	Yes (synchronised)	Legacy; rarely needed

"Concatenating in a loop with `+` creates a new String every iteration, so a thousand-iteration loop creates a thousand throwaway objects. `StringBuilder` mutates one buffer, so it is $O(n)$ instead of $O(n^2)$."

Q14. `==` vs `.equals()`

`==` compares **references** — are these the same object in memory.

`.equals()` compares **values** — as defined by the class.

```
String a = "hello";           // goes into the string pool
String b = "hello";           // reuses the SAME pooled object
String c = new String("hello"); // forces a NEW object on the heap

a == b           // true  - same pooled reference
a == c           // false - different objects
a.equals(c)      // true  - same characters
```

This is a favourite quick question. Learn the three lines.

Say this.

"`==` compares references — is this the same object in memory. `.equals()` compares values, as the class defines them. For Strings this catches people out because literals are pooled, so two identical literals are the same object and `==` happens to be true. But `new String("abc")` forces a separate object and `==` is false, while `.equals()` is still true. So I never use `==` for value comparison — it works by accident until it does not."

Q15. Stack vs Heap

- **Stack** — one per thread. Holds method calls, local variables and references. Freed automatically when the method returns.
- **Heap** — shared by all threads. Holds all objects. Managed by the garbage collector.

`StackOverflowError` means runaway recursion. `OutOfMemoryError` means the heap is full, usually a leak or an unbounded collection.

Say this.

"The stack is per thread and holds method frames, local variables and references. It is cleaned up automatically when a method returns. The heap is shared across threads and holds all the objects, and the garbage collector manages it. A `StackOverflowError` means runaway recursion; an `OutOfMemoryError` means the heap filled up, which in my experience usually means something unbounded — loading an entire table into a list instead of paginating it."

Part D · Exceptions

Q16. Checked vs unchecked ●

```
Throwable
├── Error                – JVM level, do not catch (OutOfMemoryError)
└── Exception
    ├── RuntimeException – UNCHECKED (NullPointerException, IllegalArgumentException)
    └── everything else – CHECKED (IOException, SQLException)
```

- **Checked:** the compiler forces you to catch it or declare `throws`. Represents recoverable, expected conditions — a missing file, a network failure.
- **Unchecked:** extends `RuntimeException`. Usually a programming bug. No compiler enforcement.

Say this:

"In application code I mostly use unchecked exceptions and a custom hierarchy — a `TransactionNotFoundException` extending `RuntimeException` — because forcing every caller to catch and rethrow adds noise. I handle them centrally in Spring with `@ControllerAdvice` so the API returns a consistent error shape. It also matters for `@Transactional`, because by default Spring only rolls back on unchecked exceptions."

That last sentence connects two topics and sounds like real experience. Use it.

Real-time example. In the recall APIs I had `TransactionNotFoundException` and `InvalidRecallStateException`, both unchecked, both extending a common `BusinessException`. `@RestControllerAdvice` mapped them to 404 and 409 respectively, so the UI team always got the same error shape. Making them unchecked also mattered for `@Transactional`, because a checked exception would have **committed** the transaction instead of rolling it back.

Q17. try-with-resources and finally

```
try (Connection conn = dataSource.getConnection()) {
    // conn.close() is called automatically, even if an exception is thrown
}
```

Anything implementing `AutoCloseable` works. This replaced the old `finally { if (conn != null) conn.close(); }` pattern.

Trap: "Does `finally` always run?" → Almost always. Not if the JVM exits (`System.exit()`) or the thread is killed. And if `finally` contains a `return`, it overrides the one in `try` — which is why you never return from `finally`.

Say this.

"Try-with-resources closes anything that implements `AutoCloseable` automatically, even if an exception is thrown, so I do not need a `finally` block that null-checks and closes. It replaced a pattern that people got wrong constantly — forgetting to close in one branch, or throwing a second exception inside `finally` that hid the original one. With JPA and Spring I rarely manage connections myself, but for file and stream handling it is what I use."

Q18. `throw` VS `throws` VS `final` VS `finally` VS `finalize`

- `throw` — actually raise an exception now.
- `throws` — declare that this method may raise one.
- `final` — a constant, a non-overridable method, or a non-extendable class.
- `finally` — the block that runs whatever happens.
- `finalize()` — a deprecated, unreliable method the GC used to call. Never use it.

Say this.

"`throw` raises an exception now, `throws` declares that a method might. `final` means a constant, a method that cannot be overridden or a class that cannot be extended. `finally` is the block that runs whatever happens. And `finalize()` is a deprecated method the garbage collector used to call — it is unreliable and should not be used; try-with-resources or an explicit close is the correct approach."

✓ Check yourself before moving on

1. Explain **HashMap put and get end to end** — hashing, buckets, collisions, treeification, resizing. Out loud, without notes.
2. Explain the `equals` / `hashCode` contract and what breaks if you ignore it.
3. Overloading vs overriding, with an example of each.
4. Abstract class vs interface, and when you would pick each.
5. Explain the three lines of the `==` vs `.equals()` String example.

02 · Java 8 and Functional Programming

Time needed: 60 minutes

Your resume says "Java 8+", so this **will** be tested. Interviewers usually ask you to explain streams and then write one on the spot.

First, the context: what problem did Java 8 solve?

Before Java 8, processing a collection meant writing *how* to loop:

```
// imperative - you describe every step
List<String> names = new ArrayList<>();
for (Txn t : txns) {
    if (t.getAmount() > 1000) {
        names.add(t.getCustomerName().toUpperCase());
    }
}
Collections.sort(names);
```

Java 8 let you describe *what* you want instead:

```
// declarative - the intent is visible in one read
List<String> names = txns.stream()
    .filter(t -> t.getAmount() > 1000)
    .map(t -> t.getCustomerName().toUpperCase())
    .sorted()
    .collect(Collectors.toList());
```

Same result, but the second reads as a sentence: filter, map, sort, collect. That is the point — less boilerplate and clearer intent.

Q1. What is a lambda? ●

A short way to write an implementation of an interface that has exactly one method.

```
// before Java 8
Runnable r = new Runnable() {
    public void run() { System.out.println("running"); }
};

// with a lambda
Runnable r = () -> System.out.println("running");
```

The compiler knows `Runnable` has one method, so you only supply the body.

Say this.

"A lambda is a compact way to supply the implementation of an interface that has one method. Before Java 8 I would write an anonymous inner class of five lines to pass behaviour into a method; a lambda is the same thing with the boilerplate removed. It is what makes the Streams API readable — `filter(t -> t.getAmount() > 1000)` says what it does in one line."

Q2. What is a functional interface? •

An interface with **exactly one abstract method**. That is what makes it a valid target for a lambda. `@FunctionalInterface` is optional but makes the compiler enforce it.

The four you must know:

Interface	Method	Takes	Returns	Used by
<code>Predicate<T></code>	<code>test(T)</code>	T	boolean	<code>filter</code>
<code>Function<T,R></code>	<code>apply(T)</code>	T	R	<code>map</code>
<code>Consumer<T></code>	<code>accept(T)</code>	T	void	<code>forEach</code>
<code>Supplier<T></code>	<code>get()</code>	nothing	T	<code>orElseGet</code> , lazy values

```
Predicate<Txn> isLarge = t -> t.getAmount() > 1000;
Function<Txn, String> name = t -> t.getCustomerName();
Consumer<Txn> log = t -> System.out.println(t);
Supplier<Txn> empty = () -> new Txn();
```

Also worth naming: `BiFunction<T,U,R>` (two arguments), `UnaryOperator<T>` (T to T), and `BinaryOperator<T>` (two Ts to one T, used by `reduce`).

Say this.

"An interface with exactly one abstract method. That single method is what lets the compiler match a lambda to it. `@FunctionalInterface` is optional but I add it, because it makes the compiler fail if someone later adds a second abstract method and silently breaks every lambda using it. The four I use most are `Predicate` for filtering, `Function` for mapping, `Consumer` for side effects and `Supplier` for a lazily produced value."

Q3. Method references

Shorthand for a lambda that just calls one method.

```
.map(t -> t.getCustomerName()) → .map(Txn::getCustomerName)
.forEach(x -> System.out.println(x)) → .forEach(System.out::println)
```

Say this.

"A method reference is a shorter form of a lambda that does nothing but call one method. `t -> t.getCustomerName()` becomes `Txn::getCustomerName`. It is purely readability — the compiler produces the same thing — but in a stream chain it removes noise and leaves only the intent."

Q4. ●● Streams — intermediate vs terminal operations

This distinction is asked constantly.

Intermediate operations return another Stream and are **lazy** — they do nothing until a terminal operation runs. `filter`, `map`, `flatMap`, `sorted`, `distinct`, `limit`, `skip`, `peek`.

Terminal operations produce a result and **trigger** the whole pipeline. `collect`, `forEach`, `reduce`, `count`, `min`, `max`, `anyMatch`, `allMatch`, `findFirst`.

```
txns.stream().filter(t -> t.getAmount() > 1000); // NOTHING happens - no terminal op
```

Why laziness matters — the answer that impresses:

"Because intermediate operations are lazy, the stream can fuse them into a single pass and stop early. `stream().filter(...).findFirst()` does not filter the whole list and then take the first — it stops at the first match. With `limit(10)` on a million records, only enough elements are processed to produce ten."

A stream can only be consumed once. Reusing one throws

`IllegalStateException: stream has already been operated upon or closed`.

Q5. The stream methods you must be able to write ●

```
// filter + map + collect
List<String> big = txns.stream()
    .filter(t -> t.getAmount() > 1000)
    .map(Txn::getCustomerName)
    .collect(Collectors.toList());

// sum with reduce, and the shorter specialised form
double total = txns.stream().mapToDouble(Txn::getAmount).sum();
double same = txns.stream().map(Txn::getAmount).reduce(0.0, Double::sum);

// GROUP BY - very commonly asked
Map<String, List<Txn>> byStatus =
    txns.stream().collect(Collectors.groupingBy(Txn::getStatus));

// group and count
Map<String, Long> countByStatus =
    txns.stream().collect(Collectors.groupingBy(Txn::getStatus, Collectors.counting()));

// group and sum
Map<String, Double> totalByStatus = txns.stream()
    .collect(Collectors.groupingBy(Txn::getStatus,
        Collectors.summingDouble(Txn::getAmount)));

// sort by a field, descending
txns.stream().sorted(Comparator.comparing(Txn::getAmount).reversed())

// flatMap - flatten nested collections into one stream
List<Item> all = orders.stream()
    .flatMap(o -> o.getItems().stream())
    .collect(Collectors.toList());
```

Real-time example — where this replaced real code. The report summary needed totals per status: how many recalls are pending, approved and rejected, and the total value of each. That was originally nested loops with counters and a map being built by hand. One `groupingBy` with `summingDouble` replaced about twenty lines, and the intent is now visible in one read.

⚠ But note the boundary: **do this in SQL when the data is in the database**. Streaming a million rows into memory to group them is much slower than a `GROUP BY`. Streams are for data you already have in hand.

That caveat is worth saying out loud — it shows you know where the work should happen.

● `map` vs `flatMap` is a guaranteed question:

"`map` transforms one element into one element. `flatMap` transforms one element into a **stream** of elements and then flattens them all into a single stream. If each order has a list of items, `map` gives me a stream of lists, and `flatMap` gives me a stream of items."

Real-time example. A recall request has a list of approvals. To get every approver across a page of requests, `map` would give me a stream of *lists* of approvals; `flatMap` gives me a single stream of approvals I can then filter and count.

Q6. Optional ●

Introduced to make "this might be absent" explicit in the type, instead of returning `null` and hoping the caller checks.

```
Optional<Txn> found = repository.findById(id);

// good
String name = found.map(Txn::getCustomerName).orElse("Unknown");
found.ifPresent(t -> log.info("Found {} ", t.getId()));
Txn txn = found.orElseThrow(() -> new TransactionNotFoundException(id));

// bad - this is just a null check with extra steps
if (found.isPresent()) { Txn t = found.get(); }
```

"Optional makes the possibility of absence part of the method signature, so the caller cannot ignore it by accident. I use `map`, `orElse` and `orElseThrow` rather than `isPresent` and `get`, because that defeats the purpose. Spring Data returns `Optional` from `findById` for exactly this reason."

Real-time example. `txnRepository.findById(id)` returns `Optional<Transaction>`. In the recall service that becomes `.orElseThrow(() -> new TransactionNotFoundException(id))`, which the `@RestControllerAdvice` turns into a clean 404. Before `Optional`, that same code path returned `null` and produced a `NullPointerException` and a 500 — which tells the caller nothing useful.

orElse vs orElseGet — a good detail: `orElse(buildDefault())` evaluates the argument **always**, even when a value is present. `orElseGet(() -> buildDefault())` only runs it when needed. It matters when the default is expensive.

Q7. Parallel streams — and when not to use them

```
list.parallelStream().filter(...).collect(...);
```

"It splits the work across the fork-join pool. It only helps with a large dataset and a genuinely CPU-bound operation with no shared mutable state. For small collections the coordination overhead makes it slower, and for I/O-bound work like database calls it is the wrong tool. I would measure before using it, not assume."

Q8. Other Java 8 features worth naming

- **Default and static methods in interfaces** — let you add a method to an interface without breaking every existing implementation.
- **java.time** — `LocalDate`, `LocalDateTime`, `Duration`. Immutable and thread safe, unlike the old `Date` and `Calendar`.
- **CompletableFuture** — asynchronous composition, useful for calling several services in parallel.

Say this.

"The three I actually use beyond streams. **Default methods** in interfaces, which let a library add a method without breaking every existing implementation — that is how `Collection` gained `stream()` without breaking the world. **java.time**, because the old `Date` and `Calendar` were mutable and not thread safe, and `LocalDate` and `LocalDateTime` are neither. And **CompletableFuture** when I need to call two independent services in parallel instead of one after the other."

Q9. Later versions, in one line each

If asked "what have you used beyond Java 8":

- **Java 11** — `var` for local variables, `String.isBlank()`, `strip()`, `lines()`.
- **Java 17 (LTS)** — records, sealed classes, switch expressions, text blocks.
- A **record** is a compact immutable data carrier: `record TxnDto(String id, double amount) {}` generates the constructor, getters, `equals`, `hashCode` and `toString`. Very useful for DTOs.

⚠ **The JD says "Java 8/11/17+", and your resume says Java 8.** So expect this question. Being able to name what changed in each LTS is enough — nobody expects production experience with all three.

Java 11 (LTS)

```
var list = new ArrayList<String>();           // local variable type inference
" hi ".strip();                             // Unicode-aware trim()
"".isBlank();                               // blank vs isEmpty()
"a\nb".lines();                             // stream of lines
String.join(", ", list);
list.toArray(String[]::new);
Files.readString(path);                     // read a whole file in one call
```

Java 17 (LTS)

```

// record - an immutable data carrier. Constructor, getters, equals,
// hashCode and toString are all generated.
record CardApplicationDto(Long id, String reference, BigDecimal limit) { }

// text block
String query = """
    SELECT id, status FROM card_application
    WHERE status = ?
    """;

// switch expression - returns a value, no fall-through, must be exhaustive
String label = switch (status) {
    case PENDING, IN_REVIEW -> "Awaiting action";
    case APPROVED             -> "Approved";
    case REJECTED             -> "Rejected";
};

// pattern matching for instanceof
if (obj instanceof CardApplication app && app.isPending()) {
    app.approve();                // no cast needed
}

// sealed - restricts who may extend, so a switch over them can be exhaustive
sealed interface Event permits Approved, Rejected { }

```

Say this — honestly.

"My production work has been Java 8, which is what the platform ran on. I have kept up with the LTS releases: Java 11 added `var`, the String utilities and `Files.readString`, and Java 17 added records, text blocks, switch expressions, pattern matching for `instanceof` and sealed classes.

Records are the one I would most want in day-to-day work — a DTO goes from forty lines of constructor, getters, `equals` and `hashCode` to a single line, and it is immutable by default. On the platform we used Lombok to solve the same problem, and a record is the language doing it natively."

That Lombok connection is a strong detail — it is on your resume, and it shows you understand *what problem* records solve rather than just naming the feature.

Do not claim production experience with a version you have not used. Knowing what changed and why is enough at this level.

✓ Check yourself before moving on

1. Write a stream that groups transactions by status and sums the amount in each group. From memory.

2. Explain intermediate vs terminal, and why laziness lets a stream stop early.
3. Explain `map` vs `flatMap` in two sentences.
4. Name the four core functional interfaces and which stream method uses each.

03 · Spring Boot

Time needed: 90 minutes

"Java → Spring Boot" is listed as the **primary skill** on the job description. Along with Core Java, this is where most of your technical round will be spent.

| First, the context: what problem does Spring solve?

The problem: tight coupling

Without Spring, an object creates the things it depends on:

```
public class TransactionService {  
    private final EmailNotifier notifier = new EmailNotifier();    // hard-wired  
}
```

Two problems. To switch to SMS you must edit this class. And in a unit test you cannot replace the notifier with a fake, so your test sends real emails.

The solution: Inversion of Control

The object stops creating its dependencies and **receives** them instead:

```
@Service  
public class TransactionService {  
    private final Notifier notifier;                                // an interface  
  
    public TransactionService(Notifier notifier) {                // handed in from outside  
        this.notifier = notifier;  
    }  
}
```

Now the class depends on an interface, not a concrete class. Something else decides which implementation to supply. That "something else" is the Spring container.

Inversion of Control (IoC) is the principle: control of creating objects is inverted, moving from your class to the framework.

Dependency Injection (DI) is how Spring implements it: it constructs objects and passes their dependencies in.

Say it in one line:

"IoC is the principle that the framework, not my class, controls object creation and wiring. Dependency injection is the mechanism. The practical benefit is that my classes depend on interfaces, so I can swap implementations and inject mocks in tests without changing the class."

Real-time example. The card approval service depends on a `NotificationSender` interface. In production Spring injects the RabbitMQ implementation. In unit tests I pass a mock and assert that approval publishes exactly one message — no broker, no emails, and the test runs in milliseconds. That is only possible because the service never says `new`.

And Spring Boot on top of Spring

Plain Spring needed a lot of XML and manual configuration. Spring Boot adds:

- **Auto-configuration** — sees a database driver on the classpath and configures a `DataSource` for you.
- **Starter dependencies** — `spring-boot-starter-web` pulls in a compatible set of libraries so you are not resolving versions by hand.
- **Embedded server** — Tomcat is inside the JAR, so `java -jar app.jar` runs it. No separate application server to install.
- **Production features** — Actuator gives health checks and metrics endpoints.

"Spring Boot is Spring with opinionated defaults. It removes the configuration work so you start with a running application and override only what you need."

Q1. • The three types of injection, and which to use

```
// 1. CONSTRUCTOR injection - what you should use
@Service
public class TxnService {
    private final TxnRepository repo;
    public TxnService(TxnRepository repo) { this.repo = repo; }
}

// 2. FIELD injection - convenient, but avoid
@Autowired private TxnRepository repo;

// 3. SETTER injection - for genuinely optional dependencies
@Autowired public void setRepo(TxnRepository repo) { this.repo = repo; }
```

Why constructor injection — four reasons, give two or three:

1. The field can be `final`, so the object is **immutable** once built.
2. Dependencies **cannot be missing** — the object cannot be constructed without them.
3. **Testable without Spring** — `new TxnService(mockRepo)` just works. With field injection you need reflection or a Spring context.

4. It makes bad design **visible** — a constructor with eight parameters is obviously doing too much. Field injection hides that.

Since Spring 4.3, if a class has one constructor you do not even need `@Autowired`.

Say this.

"I use constructor injection. Three reasons. The fields can be `final`, so the object is immutable once it is built. The dependencies cannot be missing, because you cannot construct the object without them — with field injection you find out at runtime with a null pointer. And it is testable without Spring at all: `new TxnService(mockRepo)` just works, whereas field injection needs reflection or a full context.

There is a fourth benefit — it makes bad design visible. A constructor with eight parameters obviously does too much. Field injection hides that."

Q2. • The annotations you must know

Annotation	What it does
<code>@SpringBootApplication</code>	The main class. Combines the next three.
<code>@Configuration</code>	This class defines beans
<code>@EnableAutoConfiguration</code>	Turn on Boot's auto-config
<code>@ComponentScan</code>	Scan this package and below for components
<code>@Component</code>	A generic Spring-managed bean
<code>@Service</code>	A <code>@Component</code> marking business logic
<code>@Repository</code>	A <code>@Component</code> for data access — also translates DB exceptions into Spring's <code>DataAccessException</code> hierarchy
<code>@Controller</code>	Returns view names
<code>@RestController</code>	<code>@Controller</code> + <code>@ResponseBody</code> — returns JSON
<code>@Autowired</code>	Inject a dependency
<code>@Qualifier("name")</code>	Choose between multiple beans of the same type
<code>@Primary</code>	Make one bean the default when several match
<code>@Value("\${prop}")</code>	Inject a property from configuration
<code>@Bean</code>	Declare a bean from a method, usually for third-party classes
<code>@Transactional</code>	Wrap the method in a database transaction
<code>@ControllerAdvice</code>	Global exception handling across controllers

Trap: '@Component vs @Service vs @Repository — is there a functional difference?'

"Technically @Service and @Component behave identically; the difference is intent, so the layer is obvious to a reader and to tooling. @Repository is the exception — it does real work, translating vendor-specific database exceptions into Spring's `DataAccessException` hierarchy, so my service layer is not coupled to a particular database's error codes."

Q3. REST annotations

```
@RestController
@RequestMapping("/api/transactions")
public class TxnController {

    @GetMapping("/{id}")
    public TxnDto get(@PathVariable Long id) { ... }

    @GetMapping
    public List<TxnDto> search(@RequestParam(defaultValue = "0") int page) { ... }

    @PostMapping
    @ResponseStatus(HttpStatus.CREATED)
    public TxnDto create(@Valid @RequestBody CreateTxnRequest req) { ... }

    @PutMapping("/{id}")
    public TxnDto replace(@PathVariable Long id, @RequestBody TxnDto dto) { ... }

    @DeleteMapping("/{id}")
    public ResponseEntity<Void> delete(@PathVariable Long id) { ... }
}
```

@PathVariable reads from the URL path. @RequestParam reads a query parameter.
@RequestBody deserialises the JSON body. @Valid triggers Bean Validation.

Say this.

"@RestController is @Controller plus @ResponseBody, so the return value is serialised to JSON rather than resolved as a view name. @RequestMapping at class level sets the base path. Then @PathVariable reads a value out of the URL path, @RequestParam reads a query parameter, and @RequestBody deserialises the JSON body into an object. I put @Valid on the request body so Bean Validation runs before the method is entered, which means invalid input never reaches my business logic."

Q3b. • Spring MVC and the DispatcherServlet — the JD names this

What happens when a request arrives, which is the question behind "do you know Spring MVC":

```
HTTP request
  → Servlet container (embedded Tomcat)
  → DispatcherServlet      the "front controller" - every request goes through it
  → HandlerMapping        which controller method handles this URL?
  → HandlerAdapter        invokes it, binding path variables, params, body
  → your @RestController method
  → HttpMessageConverter   return value -> JSON (Jackson)
  → HTTP response
```

Say this.

"Spring MVC uses the front controller pattern. A single `DispatcherServlet` receives every request and delegates: handler mapping works out which controller method matches the URL and verb, a handler adapter invokes it and binds the arguments, and on the way back a message converter serialises the return value. With `@RestController` that converter is Jackson producing JSON, which is why I do not write serialisation code myself.

The practical value of knowing this is debugging. If a request 404s, that is handler mapping. If the JSON comes back wrong or a field is missing, that is the message converter and Jackson annotations. If binding fails, that is the handler adapter. Knowing the stages tells you where to look."

That last paragraph is what turns a memorised diagram into something useful.

`@Controller` vs `@RestController` : `@Controller` returns a **view name** resolved by a view resolver — the older server-rendered model. `@RestController` is `@Controller` plus `@ResponseBody`, so the return value goes through a message converter into the response body. For a REST API you want the second.

Filter vs Interceptor, a common follow-up:

"A filter is a servlet-level concept, so it runs **before** the `DispatcherServlet` and sees every request including static resources — that is where Spring Security sits. An interceptor is a Spring MVC concept, runs inside the `DispatcherServlet`, and knows which handler was selected — so it is the right place for things like logging or timing a specific controller."

Q4. • Bean scopes

Scope	Meaning
singleton	The default. One instance for the whole application context.

Scope	Meaning
prototype	A new instance every time it is requested.
request	One per HTTP request (web only).
session	One per HTTP session (web only).

● **The follow-up:** *"Singleton beans are shared across all requests. Is that a problem?"*

"Only if the bean holds mutable state. Spring beans should be stateless — request data belongs in method parameters and local variables, not in instance fields. A singleton service with a mutable instance field is a genuine concurrency bug, because many threads share that one object."

Real-time example — a real bug this causes. If a service stores the current request's user in an instance field to avoid passing it around, two concurrent approvals will overwrite each other's value, and one approval gets attributed to the wrong approver. In a bank that is an audit failure, not just a bug. Request-scoped data belongs in method parameters, or in the `SecurityContext`, never in a field on a singleton.

That answer shows you understand *why* the default is safe, which is what they are checking.

Q5. ● Global exception handling

Nearly always asked, because it separates people who have shipped an API from people who have followed a tutorial.

```

@RestControllerAdvice
public class GlobalExceptionHandler {

    @ExceptionHandler(TransactionNotFoundException.class)
    public ResponseEntity<ErrorResponse> handleNotFound(TransactionNotFoundException ex) {
        return ResponseEntity.status(HttpStatus.NOT_FOUND)
            .body(new ErrorResponse("TXN_NOT_FOUND", ex.getMessage()));
    }

    @ExceptionHandler(MethodArgumentNotValidException.class)
    public ResponseEntity<ErrorResponse> handleValidation(MethodArgumentNotValidException ex) {
        String msg = ex.getBindingResult().getFieldErrors().stream()
            .map(e -> e.getField() + ": " + e.getDefaultMessage())
            .collect(Collectors.joining(", "));
        return ResponseEntity.badRequest().body(new ErrorResponse("VALIDATION_ERROR", msg));
    }

    @ExceptionHandler(Exception.class)
    public ResponseEntity<ErrorResponse> handleAll(Exception ex) {
        log.error("Unexpected error", ex);    // log the detail...
        return ResponseEntity.status(500)
            .body(new ErrorResponse("INTERNAL_ERROR", "Something went wrong"));
        // ...but do not leak it to the client
    }
}

```

Say why it matters:

"It keeps error handling out of every controller, so the API returns one consistent error shape. The last handler matters for security: I log the stack trace but return a generic message, because a raw stack trace tells an attacker about your framework versions and internal structure. On a banking system that is a real concern."

Real-time example. Before this was centralised, each controller had its own try/catch and every endpoint returned a slightly different error body — some with a `message`, some with an `error`, some just a stack trace. The UI team had to special-case each one. One `@RestControllerAdvice` gave every endpoint the same `{ code, message, timestamp }` shape, and their error handling became a single function.

Q6. Bean Validation

```
public class CreateTxnRequest {  
    @NotBlank(message = "Account number is required")  
    private String accountNumber;  
  
    @NotNull @DecimalMin(value = "0.01", message = "Amount must be positive")  
    private BigDecimal amount;  
  
    @Email private String notifyEmail;  
}
```

`@Valid` on the `@RequestBody` triggers it, and a failure raises `MethodArgumentNotValidException`, handled above.

⚠ Note `BigDecimal` for money, never `double`. Floating point cannot represent decimal fractions exactly, so `0.1 + 0.2` is not `0.3`. In a banking interview, using `BigDecimal` for currency is a detail that gets noticed.

Say this.

"Bean Validation with `@Valid` moves input checking to the edge of the application, so the controller rejects bad requests before any business code runs. The rules live on the DTO next to the field, which is easier to keep correct than validation scattered through a service.

The important caveat is that this is not a security control. It runs on my server, so it is trustworthy, but it is only one layer — for something like a transfer amount I would still check business rules in the service, because a validation annotation cannot know the account balance."

Real-time example. On the card request form the DTO carried `@NotBlank` on the employee ID and `@DecimalMin` on the limit. Those failures come back as a 400 with a field-level message, handled by the same `@RestControllerAdvice` as everything else, so the UI could show the error against the right field.

Q7. Configuration and profiles

```
**Say this.**
> "Profiles let one build behave differently per environment, so the artefact that passed
> SIT is the exact artefact that goes to UAT and production – only the profile changes.
> Anything environment-specific, especially credentials, comes from environment variables
> rather than being committed. And I use `@ConfigurationProperties` rather than scattering
> `@Value` annotations, because it groups related settings into one typed object that
> fails at startup if something is missing, instead of failing later at runtime."

# application.yml
spring:
  datasource:
    url: ${DB_URL}
    username: ${DB_USER}
    password: ${DB_PASSWORD}    # from environment, never committed
  ---
spring:
  config.activate.on-profile: dev
  jpa.show-sql: true
```

Run with `-Dspring.profiles.active=dev`. Profiles let one build behave differently per environment without rebuilding.

Real-time example — this is your SIT and UAT workflow. The same artefact is promoted through dev, SIT, UAT and production. Only the profile changes: different database URLs, different queue names, SQL logging on in dev and off in production. That is what makes it the *same build* being tested and released, rather than a rebuild per environment — which is the whole point of promoting a tested artefact.

```
@ConfigurationProperties(prefix = "app.recall")    // type-safe, grouped config
public class RecallProperties { private int maxDays; private String queueName; }
```

Q8. Spring Boot Actuator

Adds production endpoints: `/actuator/health` (used by load balancers and Kubernetes to decide if the instance is alive), `/actuator/metrics`, `/actuator/info`.
Say that you would **secure these**, because they expose internals.

Say this.

"Actuator exposes operational endpoints — health, metrics and info. The health endpoint is the practical one: a load balancer or container orchestrator calls it to decide whether the instance should receive traffic, and it can include the database and message broker status, so an instance that has lost its database is taken out of rotation automatically.

I would secure these endpoints, because they expose internals. On a banking system I would not have them open."

Q9. How does auto-configuration actually work?

A step above the basics, and a good one to know:

"Spring Boot ships configuration classes annotated with conditions such as `@ConditionalOnClass` and `@ConditionalOnMissingBean`. At startup it evaluates those conditions against what is on the classpath and what you have already defined. So if H2 is on the classpath and I have not declared a `DataSource`, it creates one — but the moment I define my own bean, `@ConditionalOnMissingBean` backs off and mine wins. That is why defaults never fight your own configuration."

Q10. Testing a Spring Boot application

```
@WebMvcTest(TxnController.class)           // controller layer only, fast
class TxnControllerTest {
    @Autowired MockMvc mockMvc;
    @MockBean TxnService service;           // replaces the real bean

    @Test void returns404WhenMissing() throws Exception {
        when(service.findById(1L)).thenReturn(new TransactionNotFoundException(1L));
        mockMvc.perform(get("/api/transactions/1")).andExpect(status().isNotFound());
    }
}
```

- `@SpringBootTest` loads the whole context — integration tests, slower.
- `@WebMvcTest` / `@DataJpaTest` load one slice — much faster.
- `@MockBean` swaps a bean for a Mockito mock.

Your resume mentions unit tests for state-transition validation, so be ready to say what you actually tested and why.

Say this.

"I write mostly slice tests rather than loading the whole context, because they are much faster and a fast suite is one people actually run. `@WebMvcTest` for the controller layer with the service mocked, `@DataJpaTest` for repository queries against an in-memory database, and plain JUnit with Mockito for service logic, which needs no Spring at all. `@SpringBootTest` I keep for a small number of genuine end-to-end paths.

What I test first is the state transitions — that a rejected request cannot be approved, that an already-approved one cannot be approved twice. Those are the rules that cost money if they break."

✓ Check yourself before moving on

1. Explain IoC and DI, and why constructor injection is preferred — three reasons.
2. Explain what a singleton bean means for thread safety.
3. Write a `@RestControllerAdvice` handler from memory.
4. Explain how auto-configuration decides what to configure.

04 · JPA, Hibernate and the N+1 Problem

Time needed: 75 minutes

This is the highest-value technical file for a Spring Boot backend interview. You built Spring Data JPA / PostgreSQL backends for two applications, so JPA questions are guaranteed and they will go deeper than the annotations.

N+1 is the classic question here. **Q3 has a template to fill with your own example** — and an honest fallback if you have not personally fixed one. Do not borrow a number from anywhere; a made-up metric collapses on the first follow-up.

| First, the context: what is JPA and what is Hibernate?

Your database stores **rows in tables**. Your Java code works with **objects**. Someone has to translate between them. Doing it by hand means endless JDBC boilerplate: open a connection, build SQL, loop the ResultSet, map each column to a field, close everything.

ORM (Object-Relational Mapping) automates that translation.

- **JPA** is the *specification* — a set of interfaces and annotations. It is a standard, not an implementation.
- **Hibernate** is the most common *implementation* of that specification.
- **Spring Data JPA** is a layer above both that generates repository implementations from method names, so you rarely write queries at all.

"JPA is the specification, Hibernate is the implementation, and Spring Data JPA is the abstraction on top that generates the repository code. Coding against JPA interfaces means I could switch the provider without rewriting my entities."

Q1. Entity mapping basics

```
@Entity
@Table(name = "transactions")
public class Transaction {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    @Column(nullable = false, length = 20)
    private String accountNumber;

    private BigDecimal amount;

    @Enumerated(EnumType.STRING)    // store "PENDING", not 0 - ordinals break on reorder
    private TxnStatus status;

    @ManyToOne(fetch = FetchType.LAZY)
    @JoinColumn(name = "customer_id")
    private Customer customer;
}
```

⚠ `@Enumerated(EnumType.STRING)` is worth mentioning. The default is `ORDINAL`, which stores the enum's position as a number — so inserting a new value in the middle of the enum silently corrupts every existing row. In banking data that is a serious bug.

Say this.

"`@Entity` marks the class as mapped to a table and `@Id` is the primary key, with `@GeneratedValue` telling Hibernate the database generates it. `@Column` constrains the mapping, and `@ManyToOne` with `@JoinColumn` maps the foreign key.

Two details I always set explicitly. `@Enumerated(EnumType.STRING)`, because the default stores the enum's numeric position — so if someone inserts a new value in the middle of the enum, every existing row silently means something different. And `BigDecimal` for money, never `double`, because floating point cannot represent decimal fractions exactly and the rounding errors accumulate."

Q2. • Lazy vs eager loading

- **EAGER** — the related entity is loaded immediately, with the parent.
- **LAZY** — a proxy is put in place, and the real data is only fetched when you first access it.

The defaults, which they may ask you to recite:

Relationship	Default
@OneToOne	EAGER
@ManyToOne	EAGER
@OneToMany	LAZY
@ManyToMany	LAZY

Say this:

"I set @ManyToOne to LAZY explicitly, because the default is EAGER and that means every time I load a transaction I also load its customer, even when I only wanted the amount. On a list of a thousand transactions that is a thousand unnecessary joins or queries."

Real-time example. Transaction has a @ManyToOne to Customer and to Account.

The recall list screen shows only reference number, amount and status — no customer data at all. With the EAGER default, every single row still dragged its customer and account across, which is a large part of why that page was slow. Setting them LAZY meant the list query loaded only what the screen actually displays.

The catch with LAZY: if you access a lazy field after the persistence context is closed, you get `LazyInitializationException`. The correct fixes are to fetch what you need in the query (a join fetch), or map to a DTO inside the transaction. The wrong fix is `spring.jpa.open-in-view=true`, which keeps the session open for the whole request and hides the problem while making it worse.

Q3. ●● The N+1 problem — your story

This is the most important answer in this file. Be able to explain it in 60 seconds.

What it is

```
List<Transaction> txns = txnRepository.findAll();           // 1 query - gets 100 transactions

for (Transaction t : txns) {
    System.out.println(t.getCustomer().getName());         // 1 query EACH - 100 more
}
```

One query to get the list, then N more queries — one per row — to load each row's relation. Hence **N+1**. With 100 transactions that is **101 database round trips** instead of one or two.

Why it is so slow

Each round trip has network latency. Even at 5ms each, 100 extra queries is half a second of pure waiting, and it grows linearly with the data. This is why a report that

was fine in testing crawls in production — the row count went up.

How to find it

"Turn on `spring.jpa.show-sql` or set the Hibernate SQL logger to DEBUG, then load the page and count the queries. If you see the same query repeating with a different ID, that is N+1. I also looked at execution plans to confirm which queries were actually costly."

How to fix it — know all four

1. JOIN FETCH — the usual fix.

```
@Query("SELECT t FROM Transaction t JOIN FETCH t.customer WHERE t.status = :status")
List<Transaction> findByStatusWithCustomer(@Param("status") TxnStatus status);
```

One query, everything loaded.

2. `@EntityGraph` — the declarative version.

```
@EntityGraph(attributePaths = {"customer", "account"})
List<Transaction> findByStatus(TxnStatus status);
```

3. Batch fetching — turns N queries into N/batch queries.

```
spring.jpa.properties.hibernate.default_batch_fetch_size: 25
```

4. A DTO projection — the best option for a read-only report.

```
@Query("SELECT new com.bank.dto.TxnReportDto(t.id, t.amount, c.name) " +
        "FROM Transaction t JOIN t.customer c WHERE t.status = :status")
List<TxnReportDto> findReportData(@Param("status") TxnStatus status);
```

This selects only the columns you need, avoids loading full entities into the persistence context, and skips dirty-checking entirely.

⚠ **The trap with JOIN FETCH:** fetching **two** collections in one query produces a cartesian product. Fetch one collection at a time, or use batch fetching for the second. Mentioning this unprompted marks you out.

● Your version of the answer — decide this before Saturday

If you did fix one, use this frame with something real:

"[Where it showed up.] [How I found it — SQL logging, and I saw the same query repeating once per row.] [What I changed — join fetch, `@EntityGraph`, batch fetching or a DTO projection.] [The effect — a number if you measured one, otherwise 'it went from one query per row to a single query'.]"

If you have not, answer on the mechanism. This scores well and cannot be broken:

"I have not had to fix one myself, but I know what it is and how I would find it. If an endpoint were slow I would turn on SQL logging and count the queries — the same query repeating once per row is $N+1$. I would fix it with a join fetch or an `@EntityGraph`, or a DTO projection for a read-only screen, then re-check the query count."

What you genuinely can claim, and should

From your own work at Standard Bank:

- **The Camunda workflow on Business Card**, which cut card processing time by **60%** — that is your headline number and it is a *workflow* story, not a query-tuning one. Do not blur the two.
- **The AOP audit system** writing **50,000+ transactions a day** to PostgreSQL. That is a volume story and a clean-architecture story. See file 11.
- **JasperReports** generating **1,000+ monthly reports**, and **FileNet** for document storage.
- **Liquibase** for versioned schema migrations and **Lombok** to cut persistence boilerplate — both worth mentioning here because they are JPA-adjacent and few candidates bring them up.

Q4. • `@Transactional`

```
@Transactional
public void recallTransaction(Long id) {
    Transaction txn = repo.findById(id).orElseThrow();
    txn.setStatus(RECALLED);           // no save() needed - dirty checking
    auditService.record(txn);
    // commit on normal return, rollback on a RuntimeException
}
```

The three things they ask:

1. What does it actually do? Spring creates a **proxy** around the bean. The proxy opens a transaction before the method, commits after it returns, and rolls back on failure.

2. When does it roll back? By default on **unchecked** exceptions (`RuntimeException` and `Error`) only. A checked exception **commits** unless you say `@Transactional(rollbackFor = Exception.class)`. This surprises people and is a favourite question.

3. ● Why does `@Transactional` not work when you call the method from inside the same class?

"Because it works through a proxy. An external caller goes through the proxy, so the transaction starts. But an internal call — `this.otherMethod()` — bypasses the proxy entirely and calls the real method directly, so no transaction is started. The fix is to move that method into a separate bean, so the call goes through the proxy again."

That is a senior-level answer and interviewers remember it.

Propagation, briefly: `REQUIRED` is the default — join the existing transaction or start one. `REQUIRES_NEW` suspends the current one and starts an independent one, which is what you want for audit logging that must persist even if the main work rolls back.

Real-time example — the clearest use of `REQUIRES_NEW`. A recall attempt fails validation halfway through and the main transaction rolls back, which is correct: no partial state is written. But the **audit record of the attempt must survive**, because compliance needs to know someone tried. If the audit insert joins the main transaction it is rolled back with everything else. `REQUIRES_NEW` on the audit method gives it its own transaction that commits independently.

In a banking system that is not a nicety — an audit trail that disappears when something fails is worse than no audit trail, because you believe it.

Isolation levels, in one line each:

- `READ_UNCOMMITTED` — can see uncommitted data (dirty reads). Rarely used.
- `READ_COMMITTED` — only committed data. The usual default in PostgreSQL and Oracle.
- `REPEATABLE_READ` — the same row reads the same within a transaction. MySQL's default.
- `SERIALIZABLE` — full isolation, slowest.

Real-time example — why isolation matters here. Two approvers open the same card request and click Approve at the same moment. Both transactions read the status as PENDING, both see it as approvable, and both write APPROVED — so the request is approved twice and two approval records exist. That is a **lost update**.

The fixes are optimistic locking with an `@Version` column, so the second write fails and can be retried, or a `SELECT ... FOR UPDATE` pessimistic lock. Optimistic is usually the right choice, because genuine collisions are rare and it does not hold database locks.

Q5. Hibernate caching

- **First-level cache** — the persistence context. **Always on**, scoped to one transaction. Loading the same entity twice in one transaction hits the database once.
- **Second-level cache** — optional, shared across transactions, needs a provider like Ehcache or Hazelcast. Good for reference data that rarely changes, such as a list of branch codes. Not for data that changes often.
- **Query cache** — caches query results. Must be used with the second-level cache and is easy to get wrong.

Real-time example. Branch codes, product codes and transaction-status descriptions are read on nearly every request and change perhaps twice a year. That is textbook

second-level cache material. Transactions themselves are **never** cached that way — the data changes constantly and stale financial data is worse than a slow query.

Say this.

"The first-level cache is the persistence context. It is always on and scoped to a single transaction, so loading the same entity twice in one transaction only hits the database once. That one is free and you get it whether you think about it or not.

The second-level cache is optional and shared across transactions, and it needs a provider like Ehcache. It is worth it for reference data that is read constantly and changes rarely. I would not cache transactional data that way — stale financial data is worse than a slow query, and cache invalidation across instances is where the bugs live."

Q6. Spring Data JPA repositories

```
public interface TxnRepository extends JpaRepository<Transaction, Long> {

    // Spring generates the query FROM THE METHOD NAME
    List<Transaction> findByStatusAndAmountGreaterThan(TxnStatus status, BigDecimal amt);

    Optional<Transaction> findByReferenceNumber(String ref);

    // your own JPQL when the name would get silly
    @Query("SELECT t FROM Transaction t WHERE t.createdAt BETWEEN :from AND :to")
    List<Transaction> findInRange(@Param("from") LocalDate from, @Param("to") LocalDate to);

    // native SQL when you need database-specific features
    @Query(value = "SELECT * FROM transactions WHERE MATCH(notes) AGAINST(:q)", nativeQuery = true)
    List<Transaction> search(@Param("q") String q);
}
```

Hierarchy: `Repository` → `CrudRepository` → `PagingAndSortingRepository` → `JpaRepository`. Each adds methods; `JpaRepository` adds JPA-specific ones like `flush()` and `saveAll()`.

Pagination, which you should mention for any list endpoint:

```
Page<Transaction> page = repo.findByStatus(PENDING, PageRequest.of(0, 20,
    Sort.by("createdAt").descending()));
```

Say this.

"Spring Data generates the implementation from the method name, so `findByStatusAndAmountGreaterThan` needs no code at all. I use derived queries while the name stays readable, and switch to `@Query` with JPQL once the name would become unreadable — which is usually around three conditions. Native SQL only when I need something database-specific.

The one thing I always add on a list endpoint is `Pageable`. Returning an unbounded collection is fine in testing and falls over in production when the table grows."

Q7. `save()` vs `saveAndFlush()`, and entity states

Three states: **transient** (a new object, not known to Hibernate), **persistent** (managed by the persistence context — changes are tracked automatically), **detached** (was persistent, but the context has closed).

Dirty checking is the thing to mention: for a managed entity you do not need to call `save()`. At flush time Hibernate compares the entity with its loaded snapshot and issues an UPDATE only for the fields that changed.

`saveAndFlush()` pushes the SQL to the database immediately instead of waiting for the transaction to commit. Useful when you need a generated ID within the same transaction.

Say this.

"An entity is transient when it is a plain new object Hibernate does not know about, persistent once it is managed by the persistence context, and detached after the context closes. The important consequence is **dirty checking**: while an entity is managed I do not need to call `save()` at all. I change a field, and at flush time Hibernate compares it with the snapshot it loaded and issues an update for only the changed columns.

`saveAndFlush` pushes the SQL immediately instead of waiting for the commit, which I use when I need a generated ID within the same transaction."

Q8. `findOne / getReferenceById` vs `findById`

`findById` hits the database and returns `Optional`. `getReferenceById` returns a **lazy proxy** without querying — useful when you only need to set a foreign key relationship and never read the object's fields. Touching a field on a proxy for a row that does not exist throws `EntityNotFoundException`.

Say this.

" `findById` goes to the database and returns an `Optional`. `getReferenceById` returns a lazy proxy without querying at all, which is what I want when I only need to set a foreign key — for example attaching an existing customer to a new transaction. There is no point loading the whole customer row just to write its ID.

The catch is that if the row does not exist you do not find out until you touch a field on the proxy, and then it throws `EntityNotFoundException` — possibly somewhere far from where you called it."

✓ Check yourself before moving on

1. Explain the N+1 problem, how you detect it, and **four** ways to fix it.
2. Give your N+1 answer in under 60 seconds — real example or the honest mechanism version.
3. Explain why `@Transactional` does nothing on an internal method call.
4. State the default rollback behaviour and how to change it.
5. Give the fetch-type defaults for all four relationship annotations.

05 · Microservices

Time needed: 60 minutes

Posting B lists microservices as a mandatory skill — "(Java, Spring Boot and Microservices) OR (Java, Spring Boot)" — so this is back to top priority. It is also all over your own resume: Eureka, API Gateway, JWT, and a three-service project you designed yourself. You will be asked about it whichever posting the interviewer is working from.

Both postings also say "**validate the architecture**", which is a design-review conversation. Being able to state the **cost** of a design, not just its benefit, is what that question is really testing.

If you are short of time, the highest-value parts are **Q1** (the patterns and what you give up), **Q3** (circuit breaker), **Q4** (saga) and **Q7** (how to answer the depth question honestly).

| First, the context: monolith vs microservices

The monolith

One deployable application containing everything: users, transactions, reporting, notifications. One codebase, one build, one database.

It is not a bad thing. It is simple to develop, simple to deploy, and a single transaction can span the whole system. Most applications should start here.

Where it hurts as it grows:

- One small change means redeploying everything.
- One memory leak in reporting takes down payments too.
- You cannot scale only the part that is busy — you scale the whole thing.
- A large team constantly blocks itself in one codebase.

Microservices

Split it into independent services, each owning one business capability, each with its own database, communicating over the network.

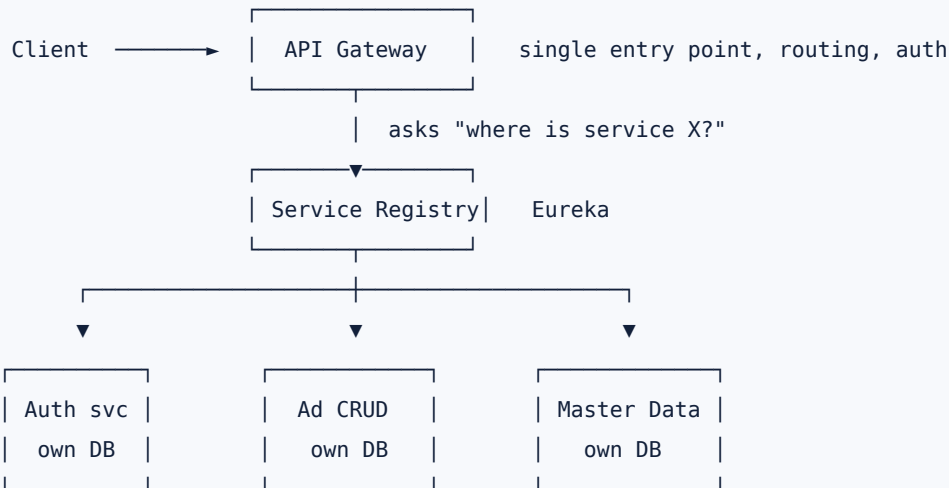
What you gain: independent deployment, independent scaling, fault isolation, and teams that can work without waiting for each other.

What you pay — say this part, because it shows judgement:

"You are trading code complexity for **operational and distributed-systems** complexity. Network calls fail, so you need retries and circuit breakers. There is no single transaction across services, so you need eventual consistency. Debugging spans several services, so you need distributed tracing and centralised logging. It is the right trade at a certain scale, but for a small application a well-structured monolith is usually the better engineering decision."

That last sentence is what separates a candidate who has *used* microservices from one who has *read* about them. Interviewers at Infosys hear the buzzword version all day.

Q1. • The core patterns — and which you used



1. Service Registry / Discovery — Eureka.

Services are on dynamic hosts and ports, and instances come and go. Hard-coding addresses does not work. Each service **registers** itself with Eureka on startup and sends heartbeats; callers **look up** the current instances by service name.

"Without discovery I would be hard-coding hostnames, which breaks the moment a service scales or moves. With Eureka a caller asks for **ad-service** and gets a live list."

2. API Gateway.

One entry point for all clients. It handles **routing**, and centralises the things you do not want to repeat in every service: authentication, rate limiting, CORS, request logging.

"Without a gateway the client would need to know every service's address, and every service would need its own copy of the auth logic. The gateway puts the cross-cutting concerns in one place."

(This is exactly the "Microservices API Management" the JD asks for. Say the words.)

3. Database per service.

Each service owns its data; no other service touches its tables. This is what makes independent deployment real — you can change your schema without coordinating with other teams.

The cost: no joins across services, and no single ACID transaction. Which leads to the next two patterns.

4. Config Server — externalised configuration for all services, so a change does not require a rebuild.

5. Circuit breaker — see Q3.

| Q2. How do services talk to each other? •

Synchronous — REST or gRPC. The caller waits for a reply.

```
@FeignClient(name = "master-data-service")    // declarative HTTP client
public interface MasterDataClient {
    @GetMapping("/categories/{id}")
    CategoryDto getCategory(@PathVariable Long id);
}
```

Simple, and you get an immediate answer. But it creates **temporal coupling**: if the callee is down, the caller fails too. Chains of synchronous calls multiply latency and failure probability.

Asynchronous — messaging over Kafka or RabbitMQ. The caller publishes an event and moves on.

"The publisher does not know or care who consumes it, and if a consumer is down the message waits in the queue. That removes the temporal coupling. The cost is eventual consistency — the effect is not immediate — and harder debugging, because the flow is no longer a straight line."

How to choose — the rule to state:

"If I need the answer to continue, synchronous. If I am telling the rest of the system that something happened, asynchronous. On the recall platform, publishing transaction state changes to Kafka for compliance logging and reporting was clearly the second kind — those consumers do not need to block the transaction from completing."

That is your project. Use it as the example.

| Q3. • Resilience — circuit breaker, retry, timeout

The problem: service A calls B, B becomes slow. A's threads pile up waiting. A runs out of threads and dies, so C which calls A dies too. One slow service takes down the system. That is a **cascading failure**.

Circuit breaker — three states, know all three:

State	Behaviour
Closed	Normal. Calls pass through, failures are counted.

State	Behaviour
Open	Failure threshold exceeded. Calls fail immediately without trying.
Half-open	After a wait, let a few test calls through. Success closes it, failure opens it again.

```
@CircuitBreaker(name = "masterData", fallbackMethod = "defaultCategory")
public CategoryDto getCategory(Long id) { return client.getCategory(id); }

public CategoryDto defaultCategory(Long id, Throwable t) {
    return CategoryDto.unknown();    // degrade gracefully instead of failing
}
```

Resilience4j is the current library; Hystrix is the older one, now retired.

"Failing fast protects both sides. The caller does not exhaust its threads waiting, and the struggling service gets breathing room to recover instead of being hammered."

Real-time example. The advertisement service calls the master data service for category names. If master data goes down without a circuit breaker, every ad request hangs on that call until it times out, the ad service's thread pool fills, and the ad service stops serving requests it could have answered perfectly well. With a breaker and a fallback, ads still render — just with a placeholder category. **Degraded is much better than down**, and that sentence is the point of the whole pattern.

Also mention **timeouts** (never make a network call without one) and **retry with exponential backoff and jitter** — and that you only retry **idempotent** operations, because retrying a payment could charge someone twice.

Q4. • Distributed transactions and the Saga pattern

The problem: placing an order must reserve stock, take payment and create a shipment — in three different services with three databases. There is no `@Transactional` that spans them.

Saga — break it into a sequence of local transactions, each publishing an event that triggers the next. If a step fails, run **compensating transactions** to undo the earlier ones.

```
Order created → Payment taken → Stock reserved → Shipment booked
                                x fails
      ← refund payment ← cancel order          (compensation)
```

Two flavours: **choreography** (services react to each other's events — simple, but the overall flow is not written down anywhere) and **orchestration** (a coordinator drives the steps — clearer and easier to debug, but it is another component to run).

Mention **eventual consistency**: the system is briefly inconsistent, and the business has to accept that. In banking that decision belongs to the business, not the developer.

Real-time example. A recall spanning services would be: mark the transaction as recalled, reverse the ledger entry, notify the customer. If the ledger reversal fails after the transaction was already marked, you cannot simply roll back across two databases — you publish a compensating event that returns the transaction to its previous state and records why. The compensation is itself an auditable business action, not a silent undo. In finance that is exactly how it has to work.

Idempotency matters here: a consumer may receive the same message twice, so processing must be safe to repeat. The usual approach is a unique message or request ID that you record and check before acting.

Say this.

"Once each service has its own database there is no distributed transaction to fall back on, so a business operation spanning services becomes a sequence of local transactions, each publishing an event that triggers the next. If a step fails, you cannot roll the earlier ones back, so you run **compensating transactions** that undo them as new business actions.

The two styles are choreography, where services react to each other's events, and orchestration, where a coordinator drives the sequence. Choreography is simpler to start but the overall flow is not written down anywhere, so I would prefer orchestration for anything with more than about three steps or anything auditable.

The thing to be explicit about with the business is that this is eventual consistency — there is a window where the system is partly updated. That is a business decision, not a technical one, and in finance it needs to be an informed one."

Q5. Observability — the question people forget

"Once a request crosses five services, a stack trace is not enough. You need three things. **Centralised logging**, so all logs land in one place — ELK or Splunk.

Distributed tracing with a correlation ID passed through every call, so you can follow one request end to end — Sleuth with Zipkin, or OpenTelemetry. And **metrics and health checks**, through Actuator and Prometheus, so you see latency and error rates per service."

Your resume mentions resolving production incidents by analysing logs, so this connects.

Q6. Other things they may ask

How do you version an API? URL versioning (`/api/v1/...`) is the most common and the most visible. The real rule is to make **additive** changes — adding an optional field is backwards compatible; removing or renaming one is not.

How do you handle security between services? The gateway validates the JWT once, then services trust the network boundary, or pass the token along and each validates it.

Service-to-service can use mTLS.

What is the Strangler Fig pattern? How you migrate a monolith gradually: put a proxy in front, move one capability at a time to a new service, and route that path to the new one. The monolith shrinks until it disappears. Nobody sensible does a big-bang rewrite.

What is a bounded context? From domain-driven design — the boundary within which a model has one consistent meaning. It is the sane way to decide **where to split services**: by business capability, not by technical layer.

"The wrong split is a 'database service' and a 'business logic service', because every change touches both. The right split is by business capability — orders, payments, notifications — so most changes stay inside one service."

How small should a service be? Not about lines of code:

"Small enough that one team can own it and understand it fully, and large enough that a typical change does not require modifying three services at once. If every feature spans multiple services, the boundaries are wrong."

Q7. • "How much microservices experience do you have?"

Your resume lists microservices under skills, and your experience is building Spring Boot services on a large enterprise banking platform. Be precise about which of those is true, because the follow-up will find out anyway.

The answer

"I should be clear about the level. On the Service Online platform I built Spring Boot services and REST APIs that other parts of the platform and a React frontend consumed, and the platform used Kafka and RabbitMQ for messaging between components. So I have worked in a distributed system and across service boundaries.

What I have not done is own the decomposition of a system into services — deciding the boundaries, standing up discovery and a gateway. I understand how those work and why they are there."

Then show the understanding, which is where you recover the ground:

"The way I think about it is that microservices buy independent deployment, independent scaling and fault isolation, and you pay with network failure, no distributed transaction, and much harder debugging. For a small system a well-structured monolith is usually the better engineering decision. The split earns its cost when teams start blocking each other or one part needs to scale differently."

Saying the costs unprompted is what separates you from candidates reciting the benefits.

Follow-ups to be ready for

- *"How would you split a monolith?"* → By **business capability**, not technical layer. Mention the **strangler fig** pattern — proxy in front, move one capability at a time.
- *"How do services communicate?"* → Synchronous REST when you need the answer to continue; messaging when you are telling the system something happened. **You have a real example** — the audit and notification paths on your platform did not need to block the main request.
- *"What happens when a downstream service is slow?"* → Timeout always, then a circuit breaker and a fallback. See Q3.

✓ Check yourself before moving on

1. Explain monolith vs microservices **including what you give up**.
2. Explain the three circuit breaker states and why failing fast helps both sides.
3. Explain the Saga pattern and compensating transactions.
4. Explain why you would use messaging instead of a REST call, with your Kafka example.
5. Give the honest Q7 answer — your level, then the trade-offs — without notes.

06 · Kafka and RabbitMQ

Time needed: 60 minutes

This is your **differentiator**. Most candidates at this level have Spring Boot and Hibernate. Far fewer have used both Kafka and RabbitMQ in production, and your resume says you used each for a *different reason*. That is the interesting part — be ready to explain the reasoning, not just name the tools.

| First, the context: why messaging at all?

The problem you actually solved

An approval endpoint that sends an email synchronously:

```
@PostMapping("/approve/{id}")
public ResponseEntity<?> approve(@PathVariable Long id) {
    cardService.approve(id);           // 200 ms
    emailService.send(...);           // 2-4 seconds, and it can fail or hang
    return ResponseEntity.ok();       // the user has been waiting the whole time
}
```

Three problems:

1. **The user waits** for something they do not care about.
2. **If the mail server is slow, the API times out** — and to the user it looks like the approval failed, even though it succeeded.
3. **Bulk approvals multiply it**. 50 approvals means 50 sequential email sends inside the request.

The fix

```
@PostMapping("/approve/{id}")
public ResponseEntity<?> approve(@PathVariable Long id) {
    cardService.approve(id);           // 200 ms
    rabbitTemplate.convertAndSend("email.queue", new EmailMessage(...)); // ~1 ms
    return ResponseEntity.ok();       // returns immediately
}
```

A separate consumer picks the message up and sends the email. The API responds in milliseconds. If the mail server is down, the message waits in the queue and is delivered when it recovers, instead of being lost.

Say it this way:

"The email is not part of what the user is waiting for, so it should not be inside the request. Moving it to RabbitMQ meant the API returned immediately, the timeouts during bulk approvals stopped, and a mail server outage no longer failed the approval — the messages just queued up until it came back."

That is exactly what your resume says you did. Make sure the *reasoning* comes out, not just the tool name.

Q1. ●● Kafka vs RabbitMQ — when do you use which?

This is the question your resume invites. You used both, so you will be asked to justify it.

	RabbitMQ	Kafka
Model	Message broker / queue	Distributed event log
After consumption	Message is removed	Message stays for its retention period
Consumers	Usually one consumer per message	Many independent consumers read the same events
Replay	No	Yes — reset the offset and re-read
Routing	Rich — exchanges, routing keys, topics	Simple — topics and partitions
Ordering	Per queue	Per partition
Throughput	High	Very high — designed for streams
Best for	Task queues, work distribution, RPC-style	Event streaming, audit trails, multiple subscribers

The one-line distinction to memorise:

"RabbitMQ is a **postbox** — a message is delivered to a worker and then it is gone. Kafka is a **ledger** — events are appended and stay there, and any number of consumers can read them at their own pace, including re-reading from the past."

● Your answer, using your own project

This is the strongest version, because it is true and it shows you chose deliberately:

"I used them for two different jobs.

RabbitMQ for the email notifications. That is a task queue: one message, one worker, do the job, and the message is done. I did not need to keep it or replay it — I needed the work off the request thread so bulk approvals stopped timing out.

Kafka for publishing transaction state changes. That is an event stream, and several downstream consumers needed it — compliance logging and report generation, and potentially more later. With Kafka each consumer group reads independently at its own pace, and because the events are retained, a consumer that was down can catch up, and compliance can replay history if it needs to rebuild. With a queue, the first consumer would take the message and the others would never see it."

That last sentence is the technical heart of it. It shows you understand the difference rather than having used whatever the team had.

Q2. RabbitMQ concepts

Producer → Exchange → (routing rules) → Queue → Consumer

The **exchange** is the piece people forget. A producer never sends to a queue directly; it sends to an exchange, which decides which queues get a copy.

Exchange type	Routes by
Direct	Exact routing key match
Topic	Wildcard pattern — <code>txn.*.approved</code>
Fanout	Copies to every bound queue, ignoring the key
Headers	Header values

```
@RabbitListener(queues = "email.queue")
public void handle(EmailMessage msg) {
    emailSender.send(msg);           // ack on normal return; requeue or DLQ on exception
}
```

Acknowledgements. The broker only removes a message once the consumer acknowledges it. If the consumer crashes mid-processing, the message is redelivered — so processing must be **idempotent**.

Dead Letter Queue (DLQ). Where messages go after repeated failures, instead of being retried forever or dropped silently. Mentioning DLQs unprompted signals production experience:

"Without a DLQ a poison message either blocks the queue or gets retried indefinitely. With one, it moves aside after N attempts, gets alerted on, and someone can inspect it without holding up everything behind it."

Q3. Kafka concepts •

Topic: transaction-events

```
└─ Partition 0 → [ msg ][ msg ][ msg ]      ← ordered within the partition
└─ Partition 1 → [ msg ][ msg ]
└─ Partition 2 → [ msg ][ msg ][ msg ]
```

Topic — a named stream of events.

Partition — a topic is split into partitions, which is how Kafka scales. Order is guaranteed **within** a partition, not across the topic.

Offset — each consumer's position in a partition. Consumers commit their offsets, so they can resume where they left off.

Consumer group — consumers in the same group **share** the partitions, one partition per consumer at a time. Consumers in **different** groups each get **all** the messages.

That last point is the mechanism behind your answer in Q1: compliance and reporting are two different consumer groups, so both see every event.

• How do you guarantee ordering in Kafka?

"By partition key. All messages with the same key go to the same partition, and order is guaranteed within a partition. So for transaction events I would key by account or transaction ID, which guarantees that everything about one transaction is processed in order, while different transactions still spread across partitions for throughput."

Delivery guarantees — know the three:

- **At most once** — commit the offset before processing. Fast, may lose messages.
- **At least once** — commit after processing. **The common default.** May duplicate, so consumers must be idempotent.
- **Exactly once** — possible with Kafka transactions, but costly and complex.

"In practice most systems run at-least-once and make consumers idempotent, because exactly-once across a distributed system is expensive. For financial events I would record a processed-message ID and check it before acting, so a redelivery is a no-op."

Replication. Each partition has a leader and followers on other brokers. If the leader dies, a follower is promoted. That is Kafka's durability story.

Q4. Quick comparison questions

"Why not just use a database table as a queue?"

"It works at small scale, and people do it. But you end up building polling, locking, retries, dead-lettering and back-pressure yourself, and polling a table is wasteful and adds latency. A broker gives you all of that, plus push delivery."

"What happens if a message fails?"

Retry with backoff, then a dead letter queue after N attempts, then alert. Never silently swallow.

"What if the same message arrives twice?"

Idempotency: a unique message ID recorded on the consumer side, or design the operation so applying it twice has the same effect as once.

"How do you not lose messages?"

Producer side: acknowledgements and retries. Broker side: persistence and replication. Consumer side: acknowledge only **after** successful processing.

✓ Check yourself before moving on

1. Kafka vs RabbitMQ in one sentence each, then **your** reason for using both.
2. Explain consumer groups and why two groups both receive every event.
3. Explain how ordering works in Kafka and how you would key transaction events.
4. Explain why at-least-once delivery means consumers must be idempotent.
5. Tell the email-timeout story with the reasoning, in 45 seconds.

07 · Spring Security and JWT

Time needed: 45 minutes

The job description names Spring Security explicitly, alongside Spring MVC and Spring Data JPA, so this is not optional. It also asks for "*secure, high-performance, scalable applications*" — security is in the responsibilities twice.

Your resume has role-based access control on the card approval flow, which is a real example to anchor to. Expect questions, and expect them to go past the annotations.

| First, the context: authentication vs authorisation

- **Authentication** — *who are you?* Checking credentials.
- **Authorisation** — *what are you allowed to do?* Checking permissions.

You authenticate once; you authorise on every request. In your Business Card system, authentication proved the user was a bank employee; authorisation decided whether that employee was a **requester**, an **approver** or an **admin**.

| Q1. How does Spring Security work?

It sits in front of your application as a **chain of servlet filters**. Every request passes through the chain before reaching a controller.

```
Request → [ filters: authenticate, authorise, CSRF, CORS ] → DispatcherServlet → Controller
```

The key filter for a token-based API is one that reads the token, validates it, and places an `Authentication` object into the `SecurityContext`. Everything downstream then knows who the caller is.

```

@Configuration
@EnableWebSecurity
public class SecurityConfig {

    @Bean
    SecurityFilterChain chain(HttpSecurity http) throws Exception {
        http
            .csrf(csrf -> csrf.disable()) // stateless API - see Q5
            .sessionManagement(s -> s.sessionCreationPolicy(SessionCreationPolicy.STATELESS))
            .authorizeHttpRequests(auth -> auth
                .requestMatchers("/api/auth/**").permitAll()
                .requestMatchers("/api/cards/approve/**").hasRole("APPROVER")
                .requestMatchers("/api/admin/**").hasRole("ADMIN")
                .anyRequest().authenticated())
            .addFilterBefore(jwtFilter, UsernamePasswordAuthenticationFilter.class);
        return http.build();
    }
}

```

Say this.

"Spring Security is a chain of servlet filters that runs before the request reaches a controller. For a token-based API the key filter reads the `Authorization` header, validates the token, and puts an `Authentication` object into the `SecurityContext` — after which everything downstream knows who the caller is and what roles they hold.

The important point is that it runs **before** my code, so an unauthenticated request never reaches business logic at all. I do not have to remember to check in each method."

Q2. • Role-based access control — your story

Two ways to enforce it. Know both.

At the URL level, as above with `hasRole(...)`.

At the method level, which is more precise:

```

@PreAuthorize("hasRole('APPROVER')")
public CardDto approve(Long id) { ... }

@PreAuthorize("hasRole('ADMIN') or #userId == authentication.principal.id")
public UserDto getUser(Long userId) { ... } // admins, or the user themselves

```

Requires `@EnableMethodSecurity`.

Your version:

"In the Business Card system there were three roles — requester, approver and admin — and the rule that mattered was that a requester must never be able to approve. I enforced it at the API layer with Spring Security rather than in the UI, because a UI check only hides the button. Anyone can call the endpoint directly with a REST client, so the authorisation has to be server side."

That last sentence is the point. In a banking interview it lands well.

⚠ **hasRole** vs **hasAuthority** : `hasRole("ADMIN")` looks for the authority `ROLE_ADMIN` — Spring adds the prefix for you. `hasAuthority("ROLE_ADMIN")` is the literal form. Mismatched prefixes are a very common bug.

Q3. • JWT — how it works

A JWT is three base64 sections separated by dots:

```
header.payload.signature

header    { "alg": "HS256", "typ": "JWT" }
payload   { "sub": "neelima", "roles": ["APPROVER"], "exp": 1735689600 }
signature HMACSHA256(base64(header) + "." + base64(payload), secret)
```

The flow

1. POST /login with credentials
2. Server verifies, signs a JWT, **returns** it
3. Client sends it **on** every request: Authorization: Bearer <token>
4. Server verifies the SIGNATURE and expiry - no **database** lookup needed
5. Request proceeds with the user's identity and roles

• Why use a JWT instead of a session?

"A session means the server stores state and every request looks it up, which is fine for one server but awkward when you scale — you need sticky sessions or a shared store like Redis. A JWT is **self-contained**: the user and roles are inside it, and the signature proves it has not been tampered with. So any instance of any service can validate it independently, with no shared session store. In a microservices setup that is the deciding factor."

• The two things people get wrong

1. A JWT is signed, not encrypted. Anyone can base64-decode the payload and read it. The signature only proves it has not been *altered*. **Never put anything sensitive in the payload** — no passwords, no account numbers.

2. You cannot easily revoke one. It is valid until it expires, so if a token is stolen it works until then. The usual answers: keep access tokens **short-lived** (15 minutes) with a longer-lived **refresh token**, and keep a small **blocklist** of revoked token IDs for logout. Say the trade-off rather than pretending it is solved.

Q4. Where should the token be stored on the client?

"An `httpOnly` cookie is safer than `localStorage`, because JavaScript cannot read an `httpOnly` cookie, so an injected script cannot steal it. Anything in `localStorage` is readable by any script on the page. The trade-off is that cookies need CSRF protection, whereas an `Authorization` header does not."

Q5. CSRF, and why it is disabled for APIs

CSRF is when another site makes the user's browser send a request to yours, relying on the browser automatically attaching the session cookie.

"It is disabled for a stateless API because the attack depends on the browser attaching credentials automatically. If the client sends an explicit `Authorization` header, another site cannot make the browser add it. If I were using cookie-based auth, CSRF protection would need to stay on."

Do not just say "we disable CSRF" — say *why*, or it looks like copied configuration.

Q6. Password storage

```
@Bean PasswordEncoder passwordEncoder() { return new BCryptPasswordEncoder(); }
```

Never store plain text, and never a plain hash like MD5 or SHA-256. BCrypt is deliberately **slow** and **salts** each password automatically, which defeats rainbow tables and makes brute force expensive.

Say this.

"Passwords are stored as a BCrypt hash, never plain text and never a plain SHA or MD5 hash. Two properties matter. BCrypt salts each password automatically, so two users with the same password get different hashes and a precomputed rainbow table is useless. And it is deliberately **slow** with a tunable work factor, so brute forcing is expensive — which is the opposite of what you want from a general-purpose hash. That slowness is the feature."

Q7. Other things worth a sentence

- **OAuth2 vs JWT** — not alternatives. OAuth2 is the authorisation *framework*; a JWT is a token *format* often used within it.

- **CORS vs CSRF** — CORS is a browser rule about which origins may read a response. CSRF is an attack. They get confused constantly.
- **Method security vs URL security** — URL rules are coarse and can be bypassed if another entry point reaches the same service method. Method-level annotations protect the method itself.

Say this if they probe the distinctions.

"OAuth2 and JWT are not alternatives — OAuth2 is an authorisation framework describing how a token is obtained, and JWT is a token format often used inside it. CORS and CSRF also get confused: CORS is a browser rule about which origins may read a response, and CSRF is an attack where another site makes the browser send an authenticated request. And URL-level security is coarse — if another entry point reaches the same service method it is unprotected, so for anything sensitive I put `@PreAuthorize` on the method itself."

✓ Check yourself before moving on

1. Explain the JWT structure and the full request flow.
2. Explain why a JWT suits microservices better than server-side sessions.
3. Explain why a JWT is not secret, and what that means for what you put in it.
4. Explain why CSRF is disabled for a stateless API.
5. Tell your RBAC story, ending on why authorisation must be server side.

08 · SQL and Query Optimisation

Time needed: 75 minutes

Your resume claims SQL tuning, indexing and execution plan analysis. That is unusual at this level and it is genuinely valuable — but it also means you **will** be tested on it. An interviewer who sees "identified and removed N+1 queries, added strategic indexing" will ask you to explain indexing properly.

Q1. Joins — know all four

```
-- INNER: only rows that match in both tables
SELECT t.id, c.name FROM transactions t
INNER JOIN customers c ON t.customer_id = c.id;

-- LEFT: every row from the left, NULLs where the right has no match
SELECT c.name, t.id FROM customers c
LEFT JOIN transactions t ON t.customer_id = c.id;    -- includes customers with no txns

-- RIGHT: the mirror image. Rarely used - people rewrite it as a LEFT JOIN.
-- FULL OUTER: everything from both sides. MySQL does not support it directly.
```

The classic question: "Find customers with no transactions."

```
SELECT c.* FROM customers c
LEFT JOIN transactions t ON t.customer_id = c.id
WHERE t.id IS NULL;
```

The `WHERE ... IS NULL` after a LEFT JOIN is the pattern. Learn it.

⚠ **A trap worth knowing:** putting a condition on the right-hand table in the `WHERE` clause of a LEFT JOIN silently turns it into an INNER JOIN, because `NULL` fails the condition. If you want to filter the joined table without losing left rows, the condition belongs in the `ON` clause.

Say this.

"An inner join returns only rows that match on both sides. A left join returns every row from the left table and nulls where the right has no match, which is what I use when the absence is the thing I care about — customers with no transactions, or requests with no approval yet. Right joins are the mirror image and people usually rewrite them as left joins for readability.

The trap is putting a condition on the right-hand table in the `WHERE` clause of a left join. Null fails the condition, so those rows are dropped and you have silently turned it back into an inner join. The condition has to go in the `ON` clause."

Q2. ●● Indexing — the answer that matters most here

What an index actually is

"An index is a separate sorted structure, usually a **B-tree**, that maps column values to row locations. Without one, the database does a **full table scan** — it reads every row to find matches, which is $O(n)$. With one, it walks the tree, which is $O(\log n)$. It is the same reason you use the index at the back of a book instead of reading every page."

When an index helps

- Columns in a `WHERE` clause
- Columns you `JOIN` on — **especially foreign keys, which are not indexed automatically in MySQL's InnoDB unless you add them**
- Columns in `ORDER BY` and `GROUP BY`

● The cost — always mention this, it is what shows real understanding

"Indexes are not free. Every `INSERT`, `UPDATE` and `DELETE` has to update every index on that table, so over-indexing slows down writes. They also take disk space. On a write-heavy table I index deliberately, not by default — the right number is the smallest set that serves the actual query patterns."

Saying only "indexes make queries faster" is the answer of someone who read it. Saying "and they slow down writes, so I add them deliberately" is the answer of someone who has tuned a real system.

Composite indexes and the left-most prefix rule

```
CREATE INDEX idx_txn_status_date ON transactions (status, created_at);
```

This index serves:

- `WHERE status = 'PENDING'` ✓
- `WHERE status = 'PENDING' AND created_at > '2026-01-01'` ✓

But **not** `WHERE created_at > '2026-01-01'` alone ✗ — because the index is sorted by `status` first. This is the **left-most prefix rule**, and it is a favourite question. Column order in a composite index matters.

Real-time example. The recall report always filters by status and then by a date range: `WHERE status = 'PENDING' AND created_at BETWEEN ? AND ?`. So the index is `(status, created_at)` in that order — the equality column first, the range column second. Reversed, as `(created_at, status)`, the index is far less effective, because the range scan happens first and status cannot narrow it. Getting that column order right was part of the 15-second to 6-second fix.

What stops an index being used

```
WHERE YEAR(created_at) = 2026      -- ✗ a function on the column defeats the index
WHERE created_at >= '2026-01-01'  -- ✓ same meaning, index usable

WHERE account_number LIKE '%1234'  -- ✗ a leading wildcard cannot use a B-tree
WHERE account_number LIKE '1234%'  -- ✓ prefix match is fine
```

Mentioning these two unprompted is a strong signal.

Clustered vs non-clustered

- **Clustered** — the table rows are physically stored in this order. One per table; in InnoDB it is the primary key.
- **Non-clustered** — a separate structure pointing back to the row.

Covering index

An index containing every column a query needs, so the database answers it from the index alone and never touches the table. Sometimes the biggest single win available.

Real-time example. A dashboard count of pending recalls per branch needs only `status` and `branch_code`. An index on `(status, branch_code)` covers it, so the database answers from the index and never reads a single table row. On a large transactions table that is the difference between milliseconds and seconds.

Q3. • EXPLAIN — how you actually diagnose

Your resume says you analysed execution plans. Be ready to describe the process.

```
EXPLAIN SELECT t.id, c.name FROM transactions t
JOIN customers c ON t.customer_id = c.id
WHERE t.status = 'PENDING';
```

What you look for:

Column	What matters
type	<code>ALL</code> = full table scan, the thing to fix. <code>ref</code> or <code>range</code> is good. <code>const</code> / <code>eq_ref</code> is best.
key	Which index was actually used. <code>NULL</code> means none.
rows	Estimated rows examined. A big number here is the problem.
Extra	<code>Using filesort</code> or <code>Using temporary</code> mean extra work. <code>Using index</code> is a covering index — good.

Say the process, not just the tool:

"I start from the slow query log or the application timings to find *which* query is slow — there is no point optimising the wrong one. Then I run `EXPLAIN` on it. If the type is `ALL` and the row count is large, it is doing a full scan, and I look at what the `WHERE` and `JOIN` columns are and whether an index exists. After adding one I re-run `EXPLAIN` to confirm it is actually being used, and re-measure the query time. Adding an index without checking the plan afterwards is guessing."

Q4. Normalisation, and when to break it

- **1NF** — atomic values, no repeating groups.
- **2NF** — 1NF, and no partial dependency on part of a composite key.
- **3NF** — 2NF, and no non-key column depending on another non-key column.

"Normalisation removes duplication, so there is one place to update a fact and no inconsistency. The cost is more joins. **Denormalisation** deliberately duplicates data to avoid expensive joins on read-heavy paths — for example storing a customer name on a report table so a reporting query does not join. It is a trade: faster reads, but now you have two copies to keep in sync."

Q5. Common query questions

Second-highest salary — asked constantly:

```
SELECT MAX(salary) FROM employees WHERE salary < (SELECT MAX(salary) FROM employees);

-- or, more generally, the Nth highest
SELECT DISTINCT salary FROM employees ORDER BY salary DESC LIMIT 1 OFFSET 1;
```

Find duplicates:

```
SELECT email, COUNT(*) FROM customers GROUP BY email HAVING COUNT(*) > 1;
```

● **WHERE vs HAVING** : `WHERE` filters **rows before** grouping. `HAVING` filters **groups after** aggregation. You cannot use an aggregate in `WHERE`.

UNION vs UNION ALL : `UNION` removes duplicates, which costs a sort. `UNION ALL` keeps everything and is faster. Use `UNION ALL` unless you actually need deduplication.

DELETE vs TRUNCATE vs DROP : `DELETE` removes rows, can have a `WHERE`, is logged and can be rolled back. `TRUNCATE` empties the table quickly, no `WHERE`, resets auto-increment. `DROP` removes the table itself.

Window functions, worth knowing one:

```
SELECT name, salary, department,  
       RANK() OVER (PARTITION BY department ORDER BY salary DESC) AS rank_in_dept  
FROM employees;
```

Unlike `GROUP BY`, a window function keeps every row and adds a computed column.

Say this when they give you a query to write.

"Let me state what I am aiming for first — I want the second highest distinct salary, so I need to handle duplicates, and decide what should happen if there is no second value."

Then write it. The pattern that scores is: restate, mention the edge case, write, then say the complexity or the index it would need. For the duplicates query, mention that `GROUP BY ... HAVING COUNT(*) > 1` is the standard shape and that on a large table the grouped column wants an index.

Q6. ACID

- **Atomicity** — all of the transaction, or none of it.
- **Consistency** — the database moves from one valid state to another; constraints hold.
- **Isolation** — concurrent transactions do not corrupt each other.
- **Durability** — once committed, it survives a crash.

The banking example makes this concrete: a transfer is a debit and a credit. Without atomicity, a crash between them destroys money. Use that example — it fits your domain.

Say this — use the banking example, it is your domain.

"A transfer is the clearest example. Debiting one account and crediting another must be **atomic** — both or neither, because a crash between them destroys money.

Consistency means constraints still hold afterwards, so the ledger still balances.

Isolation means two concurrent transfers on the same account cannot interleave and produce a wrong balance. And **durability** means once it is committed it survives a crash — the customer has been told it happened, so it has to have happened.

That is also why I would not move something like a ledger entry to eventual consistency without the business explicitly agreeing to it."

Q7. SQL vs NoSQL — you have MongoDB on your resume

"I would use a relational database when the data is structured and relational and I need ACID transactions — which is most of banking. MongoDB suits flexible or evolving schemas, document-shaped data, and cases where horizontal scaling matters more than joins. The real question is the access pattern: if I am constantly joining, that is a relational shape, and forcing it into documents means duplicating data everywhere."

Q8. Pagination at the database level

```
SELECT * FROM transactions ORDER BY created_at DESC LIMIT 20 OFFSET 0;
```

"OFFSET pagination degrades on deep pages, because the database still has to scan and discard everything before the offset. Page 5,000 is slow. **Keyset pagination** — `WHERE created_at < :lastSeen ORDER BY created_at DESC LIMIT 20` — stays fast because the index seeks straight to the position."

That is a strong detail for a reporting-heavy system, which yours was.

✓ Check yourself before moving on

1. Explain what an index is, **and its cost on writes**.
2. Explain the left-most prefix rule with the composite index example.
3. Give two things that stop an index being used.
4. Describe your EXPLAIN process end to end.
5. Write the "customers with no transactions" query from memory.
6. Explain `WHERE` vs `HAVING` .

09 · DSA Part 1 — Patterns, Arrays, Strings, Hashing

Time needed: 3 hours across two sessions. Type every solution.

✓ Every solution in files 09 and 10 has been **compiled and run against test cases**, including the edge cases. If yours does not behave the same way, it is a typo in your version, not a bug in the guide.

First, the context: what DSA level does Infosys actually test?

Set your expectations correctly, because over-preparing the wrong thing is the most common waste of time before a service-company interview.

What they ask: arrays, strings, HashMap-based counting, two pointers, basic recursion, sorting and searching, sometimes a linked list or a stack. Usually **one or two problems**, at what LeetCode would call **easy to lower-medium**.

What they almost never ask: dynamic programming, graph algorithms, tries, segment trees. If you are spending Saturday morning on DP, stop.

What they are actually checking:

1. Can you write **compiling, correct Java** without an IDE completing it for you?
2. Do you reach for the **right collection** — and can you say why?
3. Do you handle **edge cases** — null, empty, single element, duplicates?
4. Can you state the **time and space complexity** without being asked?
5. Do you **talk while you think**?

Points 4 and 5 are free marks and most candidates drop them.

How to run a coding question — the six steps

- | | |
|-----------------------|--|
| 1. RESTATE | "So I need to return the first character that appears once, and null if there isn't one ?" |
| 2. EDGE CASES | Ask before coding. Empty? Null? Case sensitive? Duplicates? |
| 3. BRUTE FORCE | Say it out loud, with its complexity. Do not write it yet. |
| 4. IMPROVE | "I can make the lookup $O(1)$ with a HashMap, so one pass." |
| 5. WRITE | Talking as you go . |
| 6. TRACE + COST | Walk one example through, then state time and space. |

Step 3 matters more than people think. Saying *"the brute force is two nested loops, $O(n^2)$ — I can do better with a map"* shows you can reason about cost. Jumping straight to

the optimal solution you memorised shows nothing.

Complexity — the table you must know cold

Operation	ArrayList	LinkedList	HashMap / HashSet	TreeMap / TreeSet
Access by index	O(1)	O(n)	—	—
Search by value	O(n)	O(n)	O(1) avg	O(log n)
Insert / delete at end	O(1)*	O(1)	O(1) avg	O(log n)
Insert / delete in middle	O(n)	O(1) at a known node	—	—
Keeps order	insertion	insertion	no	sorted

* amortised — occasionally it resizes and copies.

Sorting — `Collections.sort` / `Arrays.sort` on objects is **O(n log n)**, a stable merge sort (TimSort). `Arrays.sort` on **primitives** is a dual-pivot quicksort, O(n log n) average, and **not stable**. That distinction is a good detail to know.

Binary search — O(log n), but **only on sorted data**.

The rule of thumb to say out loud: "If I am searching inside a loop, that is O(n²), and a HashMap usually turns it into O(n) by trading space for time." That single sentence solves a large share of interview problems.

Pattern 1 ●● — Hashing for counting and lookup

The single most useful pattern. Half of all easy problems are this.

First non-repeating character

```
public static Character firstNonRepeating(String input) {
    if (input == null || input.isEmpty()) return null;

    Map<Character, Integer> counts = new LinkedHashMap<>(); // insertion order!
    for (char c : input.toCharArray()) {
        counts.merge(c, 1, Integer::sum);
    }
    for (Map.Entry<Character, Integer> e : counts.entrySet()) {
        if (e.getValue() == 1) return e.getKey();
    }
    return null;
}
```

Say: " `LinkedHashMap` , not `HashMap` , because I need insertion order to find the *first* one. Two passes, so $O(n)$ time, $O(k)$ space where k is the distinct character count."

Two sum

```
public static int[] twoSum(int[] nums, int target) {
    Map<Integer, Integer> seen = new HashMap<>(); // value -> index
    for (int i = 0; i < nums.length; i++) {
        int need = target - nums[i];
        if (seen.containsKey(need)) return new int[]{ seen.get(need), i };
        seen.put(nums[i], i);
    }
    return new int[]{};
}
```

Say: "Brute force is two nested loops, $O(n^2)$. Storing what I have already seen makes the lookup $O(1)$, so it becomes one pass — $O(n)$ time, $O(n)$ space. Classic time-for-space trade."

Find and remove duplicates

```
public static Set<Integer> findDuplicates(int[] nums) {
    Set<Integer> seen = new HashSet<>(), dupes = new HashSet<>();
    for (int n : nums) {
        if (!seen.add(n)) dupes.add(n); // add() returns false if already present
    }
    return dupes;
}

public static List<Integer> removeDuplicatesKeepingOrder(List<Integer> nums) {
    return new ArrayList<>(new LinkedHashSet<>(nums));
}
```

`seen.add(n)` returning `false` is the neat bit — one lookup instead of `contains` then `add` .

Anagram check

```
public static boolean isAnagram(String a, String b) {
    if (a == null || b == null || a.length() != b.length()) return false;

    Map<Character, Integer> counts = new HashMap<>();
    for (char c : a.toCharArray()) counts.merge(c, 1, Integer::sum);
    for (char c : b.toCharArray()) {
        Integer n = counts.get(c);
        if (n == null) return false;
        if (n == 1) counts.remove(c); else counts.put(c, n - 1);
    }
    return counts.isEmpty();
}
```

Say the trade-off: "Sorting both and comparing is simpler but $O(n \log n)$. Counting is $O(n)$ with $O(k)$ space. For a fixed alphabet I could use an `int[26]` instead of a map."

Word / element frequency

```
Map<String, Long> freq = Arrays.stream(sentence.toLowerCase().split("\\s+"))
    .collect(Collectors.groupingBy(w -> w, Collectors.counting()));

// highest frequency element
String top = Collections.max(freq.entrySet(), Map.Entry.comparingByValue()).getKey();
```

Pattern 2 ● — Two pointers

Use when the array is sorted, or when you work inward from both ends.

Palindrome

```
public static boolean isPalindrome(String s) {
    if (s == null) return false;
    String clean = s.replaceAll("[^A-Za-z0-9]", "").toLowerCase();
    int left = 0, right = clean.length() - 1;
    while (left < right) {
        if (clean.charAt(left++) != clean.charAt(right--)) return false;
    }
    return true;
}
```

Say: "Better than reversing and comparing, because it exits early on the first mismatch and uses no extra string. $O(n)$ time, $O(1)$ extra space beyond the cleaned copy."

Reverse in place

```
public static String reverse(String s) {
    char[] c = s.toCharArray();
    int i = 0, j = c.length - 1;
    while (i < j) { char t = c[i]; c[i++] = c[j]; c[j--] = t; }
    return new String(c);
}
```

They often want this **without** `new StringBuilder(s).reverse()` — know both, offer the library one first, then write the manual one if asked.

Pair with a given sum in a *sorted* array

```
public static int[] pairWithSum(int[] sorted, int target) {
    int lo = 0, hi = sorted.length - 1;
    while (lo < hi) {
        int sum = sorted[lo] + sorted[hi];
        if (sum == target) return new int[]{lo, hi};
        if (sum < target) lo++; else hi--;
    }
    return new int[]{};
}
```

Say: "O(n) time and **O(1) space** — better than the HashMap version, but only because the input is sorted."

Pattern 3 ● — Sliding window

Use for "longest / shortest / maximum subarray or substring of size k, or satisfying X".

Maximum sum of a subarray of size k

```
public static int maxSumOfSizeK(int[] nums, int k) {
    if (nums.length < k) return -1;
    int sum = 0;
    for (int i = 0; i < k; i++) sum += nums[i];

    int best = sum;
    for (int i = k; i < nums.length; i++) {
        sum += nums[i] - nums[i - k]; // slide: add new, drop old
        best = Math.max(best, sum);
    }
    return best;
}
```

Say: "Recomputing each window is $O(n \cdot k)$. Sliding reuses the previous sum, so it is $O(n)$."

Longest substring without repeating characters

```
public static int longestUnique(String s) {
    Map<Character, Integer> lastSeen = new HashMap<>();
    int best = 0, start = 0;
    for (int i = 0; i < s.length(); i++) {
        char c = s.charAt(i);
        if (lastSeen.containsKey(c) && lastSeen.get(c) >= start) {
            start = lastSeen.get(c) + 1;    // shrink the window past the duplicate
        }
        lastSeen.put(c, i);
        best = Math.max(best, i - start + 1);
    }
    return best;
}
```

This is the hardest problem you are realistically likely to get. Worth practising twice.

Pattern 4 — Array classics they actually ask

Second largest / second highest

```
public static int secondLargest(int[] nums) {
    int first = Integer.MIN_VALUE, second = Integer.MIN_VALUE;
    for (int n : nums) {
        if (n > first) { second = first; first = n; }
        else if (n > second && n != first) { second = n; }
    }
    return second;
}
```

Say: "Sorting and taking index $n-2$ is $O(n \log n)$ and breaks with duplicates. One pass is $O(n)$. The edge cases are an array shorter than two and all-equal elements."

Missing number from 1..n

```
public static int missingNumber(int[] nums, int n) {
    long expected = (long) n * (n + 1) / 2;
    long actual = 0;
    for (int x : nums) actual += x;
    return (int) (expected - actual);
}
```

Mentioning `long` to avoid overflow is a detail interviewers notice.

Rotate an array by k

```
public static void rotate(int[] nums, int k) {
    k %= nums.length;
    reverse(nums, 0, nums.length - 1);
    reverse(nums, 0, k - 1);
    reverse(nums, k, nums.length - 1);
}

private static void reverse(int[] a, int i, int j) {
    while (i < j) { int t = a[i]; a[i++] = a[j]; a[j--] = t; }
}
```

The three-reversal trick: $O(n)$ time, $O(1)$ space. Worth memorising — it looks clever and it is easy to remember.

Move zeros to the end, keeping order

```
public static void moveZeros(int[] nums) {
    int write = 0;
    for (int n : nums) if (n != 0) nums[write++] = n;
    while (write < nums.length) nums[write++] = 0;
}
```

Pattern 5 — Java 8 stream versions

Because your resume says Java 8, they may ask for the stream form.

```
// sum of even numbers
int sum = nums.stream().filter(n -> n % 2 == 0).mapToInt(Integer::intValue).sum();

// second highest distinct
Optional<Integer> second = nums.stream().distinct()
    .sorted(Comparator.reverseOrder()).skip(1).findFirst();

// group by a property
Map<String, List<Employee>> byDept = employees.stream()
    .collect(Collectors.groupingBy(Employee::getDepartment));

// highest paid per department
Map<String, Optional<Employee>> topPerDept = employees.stream()
    .collect(Collectors.groupingBy(Employee::getDepartment,
        Collectors.maxBy(Comparator.comparing(Employee::getSalary))));

// sort a map by value, descending, keeping the order
Map<String, Integer> sorted = map.entrySet().stream()
    .sorted(Map.Entry.<String, Integer>comparingByValue().reversed())
    .collect(Collectors.toMap(Map.Entry::getKey, Map.Entry::getValue,
        (a, b) -> a, LinkedHashMap::new));
```

That last one is worth practising — the four-argument `toMap` with `LinkedHashMap::new` is required to keep the sort order, and almost everyone forgets it.

⚠ **Say the boundary:** "If this data were in the database I would do the grouping in SQL with `GROUP BY` rather than streaming a million rows into memory." That shows judgement.

Java traps they enjoy in coding rounds

```
Integer a = 127, b = 127;
Integer c = 128, d = 128;
a == b           // true - Integer cache covers -128..127
c == d           // false - outside the cache
c.equals(d)      // true

String s1 = "abc", s2 = "abc", s3 = new String("abc");
s1 == s2         // true - string pool
s1 == s3         // false
s1.equals(s3)    // true

0.1 + 0.2 == 0.3 // false - floating point. Money uses BigDecimal.
```

✓ Practice plan for this file

Type these **from an empty file**, no copy-paste, talking out loud:

first non-repeating · two sum · palindrome · max sum of size k · second largest.

If you can write those five while explaining your reasoning, you are ready for this round.

10 · DSA Part 2 — Linked Lists, Stacks, Recursion, Searching, Trees

Time needed: 2 hours. Type the starred ones.

Lower probability than Part 1, but a linked list or stack question is common enough that being blank on one would hurt. Trees are less likely at this level — know the traversals and move on.

| Part A · Linked lists ●

The node, and why you write a dummy head

```
class ListNode {
    int val;
    ListNode next;
    ListNode(int val) { this.val = val; }
}
```

The single most useful trick: a **dummy head** node removes almost every special case around "what if the list is empty" or "what if I delete the first node".

★ Reverse a linked list

The most asked linked list question, by a distance.

```
public static ListNode reverse(ListNode head) {
    ListNode prev = null, curr = head;
    while (curr != null) {
        ListNode next = curr.next; // remember where we were going
        curr.next = prev;          // flip the pointer
        prev = curr;               // shuffle both forward
        curr = next;
    }
    return prev;                  // prev is the new head
}
```

Trace it out loud — three pointers, and the order of the four lines matters. If you assign `curr.next = prev` before saving `next`, you have lost the rest of the list. $O(n)$ time, $O(1)$ space.

★ Find the middle — the two-pointer trick

```
public static ListNode middle(ListNode head) {
    ListNode slow = head, fast = head;
    while (fast != null && fast.next != null) {
        slow = slow.next;
        fast = fast.next.next;    // moves twice as fast
    }
    return slow;
}
```

★ Detect a cycle — Floyd's algorithm

```
public static boolean hasCycle(ListNode head) {
    ListNode slow = head, fast = head;
    while (fast != null && fast.next != null) {
        slow = slow.next;
        fast = fast.next.next;
        if (slow == fast) return true;    // they can only meet inside a loop
    }
    return false;
}
```

Say: "A `HashSet` of visited nodes also works and is easier to explain, but it is $O(n)$ space. Floyd's is $O(1)$ — if there is a cycle, the fast pointer eventually laps the slow one."

Knowing *both* and choosing, out loud, is what scores.

Merge two sorted lists

```
public static ListNode merge(ListNode a, ListNode b) {
    ListNode dummy = new ListNode(0), tail = dummy;
    while (a != null && b != null) {
        if (a.val <= b.val) { tail.next = a; a = a.next; }
        else { tail.next = b; b = b.next; }
        tail = tail.next;
    }
    tail.next = (a != null) ? a : b;    // attach whatever is left
    return dummy.next;
}
```

The dummy head is why this has no special cases.

Remove the nth node from the end

Two pointers, n apart: advance the first n steps, then move both until the first hits the end. The second is now at the node before the one to delete. Use a dummy head so deleting

the first node is not special.

Part B • Stacks and queues •

Stack is LIFO — use `Deque<Integer> stack = new ArrayDeque<>()`.

⚠ **Do not use the legacy `Stack` class.** It extends `Vector` and is synchronised, so it is slower and its iteration order is famously confusing. Saying "`ArrayDeque`", because `java.util.Stack` is a legacy synchronised class" is a small detail that lands well.

Queue is FIFO — `Queue<Integer> q = new LinkedList<>()` or `ArrayDeque`.

★ Balanced brackets

The classic stack question.

```
public static boolean isBalanced(String s) {
    Deque<Character> stack = new ArrayDeque<>();
    Map<Character, Character> pairs = Map.of(')', '(', ']', '[', '}', '{');

    for (char c : s.toCharArray()) {
        if (c == '(' || c == '[' || c == '{') {
            stack.push(c);
        } else if (pairs.containsKey(c)) {
            if (stack.isEmpty() || stack.pop() != pairs.get(c)) return false;
        }
    }
    return stack.isEmpty();           // leftovers mean unclosed brackets
}
```

The two edge cases interviewers check: a closing bracket when the stack is **empty**, and leftovers on the stack **at the end**. Mention both.

Next greater element

For each element, the next larger one to its right. A **monotonic stack** does it in $O(n)$ instead of $O(n^2)$: push indices, and whenever the current value exceeds the value at the top of the stack, you have found that index's answer — pop and record.

Implement a queue using two stacks

Push onto `in`. To pop, if `out` is empty, drain `in` into `out` and pop from there. Each element moves at most twice, so it is **amortised $O(1)$** . Saying "amortised" is the point of the question.

Part C · Recursion •

The three rules

1. A **base case** that stops it. Without one you get `StackOverflowError`.
2. The recursive call must move **toward** the base case.
3. Trust the recursion — assume the smaller call works, and handle only this level.

Factorial and Fibonacci

```
public static long factorial(int n) {
    if (n <= 1) return 1;           // base case
    return n * factorial(n - 1);
}

// iterative Fibonacci - prefer this
public static long fib(int n) {
    if (n <= 1) return n;
    long prev = 0, curr = 1;
    for (int i = 2; i <= n; i++) {
        long next = prev + curr;
        prev = curr; curr = next;
    }
    return curr;
}

// if they insist on recursion, memoise it
public static long fibMemo(int n, Map<Integer, Long> memo) {
    if (n <= 1) return n;
    return memo.computeIfAbsent(n, k -> fibMemo(k - 1, memo) + fibMemo(k - 2, memo));
}
```

Say: "Naive recursive Fibonacci is $O(2^n)$ because it recomputes the same subproblems. Memoising makes it $O(n)$. Iterative is $O(n)$ time and $O(1)$ space, so that is what I would actually write."

That comparison is the whole reason the question is asked.

Recursion vs iteration

"Recursion is clearer for genuinely recursive structures — trees, nested directories. But each call uses a stack frame, so deep recursion risks `StackOverflowError`, and Java does not do tail-call optimisation. For a linear loop I use iteration."

Part D · Searching and sorting •

★ Binary search — write it correctly

```
public static int binarySearch(int[] sorted, int target) {
    int lo = 0, hi = sorted.length - 1;
    while (lo <= hi) {                // <= , not <
        int mid = lo + (hi - lo) / 2; // avoids integer overflow
        if (sorted[mid] == target) return mid;
        if (sorted[mid] < target) lo = mid + 1;
        else hi = mid - 1;
    }
    return -1;
}
```

Two details interviewers look for: `lo + (hi - lo) / 2` rather than `(lo + hi) / 2`, which can overflow; and `while (lo <= hi)`, because with `<` you miss the last element. $O(\log n)$, **and it only works on sorted data** — say that.

Sorting — what to know, not how to implement

Algorithm	Time	Space	Stable
Bubble / insertion	$O(n^2)$	$O(1)$	yes
Merge sort	$O(n \log n)$	$O(n)$	yes
Quick sort	$O(n \log n)$ avg, $O(n^2)$ worst	$O(\log n)$	no
Heap sort	$O(n \log n)$	$O(1)$	no

What Java actually uses: `Collections.sort` and `Arrays.sort` on **objects** use TimSort, a stable merge sort. `Arrays.sort` on **primitives** uses dual-pivot quicksort, which is not stable — but stability is meaningless for primitives, which is why that is fine.

Stable means equal elements keep their relative order. It matters when you sort by one field and then another.

```
list.sort(Comparator.comparing(Employee::getDept)
    .thenComparing(Employee::getSalary, Comparator.reverseOrder()));
```

Being able to write a chained comparator is more likely to be asked than implementing a sort.

Part E · Trees — know the traversals, do not over-invest

```
class TreeNode { int val; TreeNode left, right; }
```

The four traversals

```
// INORDER - left, node, right. On a BST this gives SORTED order.
void inorder(TreeNode n, List<Integer> out) {
    if (n == null) return;
    inorder(n.left, out); out.add(n.val); inorder(n.right, out);
}

// PREORDER - node, left, right (used to copy a tree)
// POSTORDER - left, right, node (used to delete a tree)

// LEVEL ORDER (BFS) - uses a QUEUE, not recursion
List<Integer> levelOrder(TreeNode root) {
    List<Integer> out = new ArrayList<>();
    if (root == null) return out;
    Queue<TreeNode> q = new LinkedList<>();
    q.add(root);
    while (!q.isEmpty()) {
        TreeNode n = q.poll();
        out.add(n.val);
        if (n.left != null) q.add(n.left);
        if (n.right != null) q.add(n.right);
    }
    return out;
}
```

The one fact to have ready: *"Inorder traversal of a binary search tree gives you the values in sorted order."* That is the most commonly asked tree fact.

Height, and BST search

```
int height(TreeNode n) {
    return n == null ? 0 : 1 + Math.max(height(n.left), height(n.right));
}
```

BST search is $O(\log n)$ on a balanced tree and **$O(n)$ on a degenerate one** — which is why self-balancing trees like red-black exist. That connects straight back to `TreeMap` and to `HashMap`'s treeified buckets, and making that connection unprompted sounds very good.

Do not spend Saturday morning on tree problems. Know the four traversals, the BST/inorder fact, and the height function. That is proportionate.

✓ Practice plan for this file

Type these from an empty file: **reverse a linked list · detect a cycle · balanced brackets · binary search.**

Reverse-a-linked-list and balanced brackets are the two most likely to appear. Practise tracing the reversal out loud with three pointers — explaining it is harder than writing it.

11 · Backend System Design

Time needed: 75 minutes

At this level Infosys will not ask you to design Netflix. But the JD is full of consulting language — *"problem definition, effort estimation, diagnosis, solution generation and design"* — so expect one design question, and expect them to be watching **how you think**, not whether you produce a perfect architecture.

The framework — use it for every design question

- | | |
|----------------------|---|
| 1. CLARIFY | Ask 2-3 questions. Never start designing immediately. |
| 2. SCOPE | State the core use cases you will design for . |
| 3. API | What end points, what request and response shapes. |
| 4. DATA MODEL | Tables or documents, and the key relationships. |
| 5. FLOW | Walk one request end to end . |
| 6. SCALE | What breaks first, and what you would do about it. |

Step 1 is the one candidates skip, and it is the one the JD explicitly asks for.

"Problem definition" is literally in the job description.

Clarifying questions that fit almost anything:

- "Roughly what volume are we designing for — hundreds a day or millions?"
- "Is this read-heavy or write-heavy?"
- "Does this need to be strongly consistent, or is eventual consistency acceptable?"
- "Is there an existing system I am integrating with, or is this greenfield?"

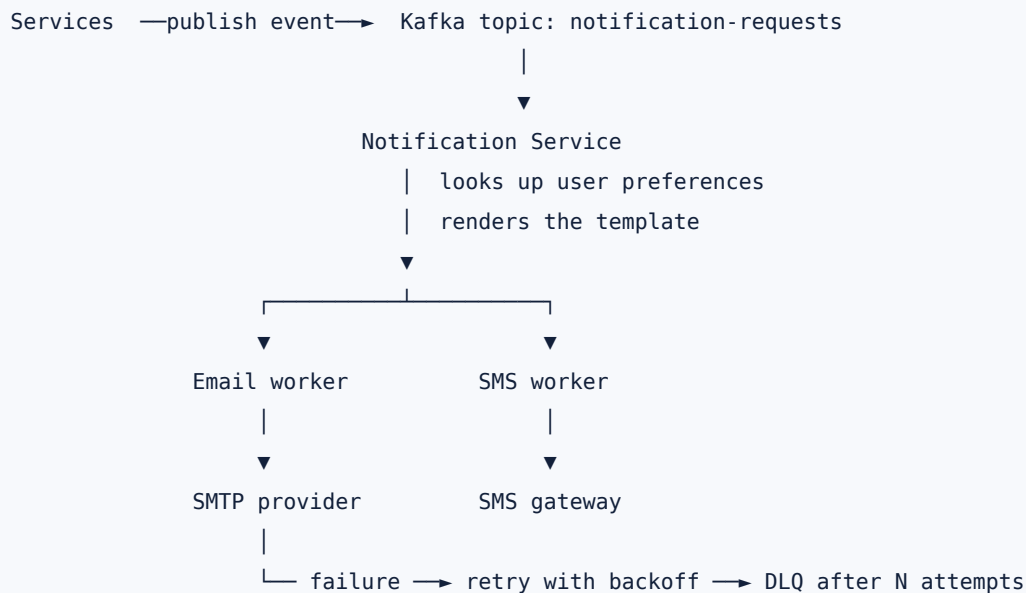
That third question is a strong one to ask in a banking context, because the answer is usually "strongly consistent for money, eventually consistent for notifications" — and saying that shows domain judgement.

Design 1 ● — A notification service

Most likely for you, because your resume already describes half of it.

Clarify: Which channels — email, SMS, push? Do we need delivery status? Are there priorities, or is everything equal? Roughly what volume?

The design:



The points to make:

- **Asynchronous, always.** The calling service must never wait for an email. This is exactly what you did with RabbitMQ.
- **Preferences and templates** live in the notification service, not scattered across callers. One place to change how something is worded.
- **Retry with exponential backoff**, then a dead letter queue, then an alert. Never silently drop a notification.
- **Idempotency**: a duplicate event must not send two emails. Key on an event ID and record what has been sent.
- **Delivery status** written back so support can answer "did the customer get it?"

Then land it: *"This is close to what I built — the async part with RabbitMQ for emails, and Kafka for publishing state changes to several consumers."*

Design 2 • — A transaction recall / approval workflow

Your actual domain. If they let you pick, pick this.

Clarify: Who can raise a recall, who approves? Is there a time limit? Does it need an audit trail? Single or multi-level approval?

Data model:

```

recall_request
  id, transaction_id, requested_by, reason, status, created_at, updated_at

recall_approval
  id, recall_request_id, approver_id, decision, comments, decided_at

audit_log
  id, entity_type, entity_id, action, actor, before_state, after_state, timestamp
  
```

The state machine — draw it, it always impresses:



The points to make:

- **Model the states explicitly** and validate transitions in one place. An approved request must never go back to draft. (*Your resume mentions unit tests for state-transition validation — say that here.*)
- **Authorisation server side.** A requester must not be able to approve their own request. Enforce in the service layer, not the UI.
- **Append-only audit log.** In banking every state change must be attributable — who, what, when, and what it was before. Never update audit rows.
- **Publish each state change as an event** for compliance and reporting consumers. That is your Kafka work.
- **Idempotency on submit**, so a double-click does not create two recall requests.

Design 3 — A URL shortener

The classic warm-up. Know it in case it comes up.

Core: `POST /shorten` takes a long URL, returns a short code. `GET /{code}` returns a **301 or 302 redirect**.

Generating the code: a base62 encoding of an auto-increment ID is simple and collision-free. Hashing the URL is an alternative but needs collision handling.

Scale points: it is extremely read-heavy, so cache the code-to-URL mapping in Redis in front of the database. The lookup must be a primary key or unique index hit. Mention 301 (permanent, browser caches it, so you lose click analytics) versus 302 (temporary, every click reaches you) — that trade-off is the interesting part.

Design 4 — "How would you design a REST API for X?"

The generic answer, which is worth having ready:

GET	/api/v1/transactions	list, with pagination and filters
GET	/api/v1/transactions/{id}	one
POST	/api/v1/transactions	create → 201 + Location header
PUT	/api/v1/transactions/{id}	full replace
PATCH	/api/v1/transactions/{id}	partial update
DELETE	/api/v1/transactions/{id}	remove → 204

The rules to state:

- **Nouns, not verbs.** `/transactions/{id}/recall` as a sub-resource, not `/recallTransaction?id=`.
- **Correct status codes.** 201 for created, 204 for no content, 400 for bad input, 401 vs 403, 404, 409 for a conflict, 422 for validation, 500 for server errors.

- **Pagination on every list endpoint.** Never return an unbounded collection.
- **Versioning** in the URL, and only additive changes within a version.
- **A consistent error shape** across every endpoint, via `@RestControllerAdvice`.
- **Idempotency** for non-safe operations, usually via a client-supplied key.
- **DTOs, never entities, at the boundary.** Returning entities leaks your schema, causes lazy-loading surprises during serialisation, and means a database change breaks your API contract.

That last point is a genuinely good one and few candidates say it.

Design 5 — "This API is slow. What do you do?"

You have lived this. Give a process, not a list of fixes.

1. MEASURE Where is the time actually **going**? Application logs and timings first, **then** the slow query log. Do not guess.
2. CATEGORISE Is it the **data**base, the application, or an external call?
 - **Database** → N+1, missing index, bad JOIN
 - Application → inefficient loop, unnecessary object creation
 - External → a synchronous call that should be async
3. FIX THE BIGGEST **ONE** FIRST
4. RE-MEASURE Prove it. Compare before and after.

Then make it concrete: *"That is exactly the process that took the report from 15 seconds to 6. The measurement step told me it was N+1 rather than the query itself, which I would not have guessed."*

The honest line that scores: *"My first assumption about the cause was wrong, which is why I measure before optimising."*

✓ Check yourself before moving on

1. Recite the six-step framework.
2. Design the notification service out loud in 3 minutes, including retries and DLQ.
3. Draw the recall state machine and say why the audit log is append-only.
4. Give five REST API design rules, including DTOs at the boundary.

12 · Talking About Your Experience

Time needed: 60 minutes, said out loud

| First, the context: you are pitching as a backend engineer now

Your resume is positioned as **Software Engineer, Backend — Java · Spring Boot · Camunda · PostgreSQL**, and the React work sits underneath as supporting. That is the right call for a Java and Spring Boot interview, and it is honest, because the backend work is substantial.

What is genuinely strong:

- **You built the backends, not just consumed them.** Spring Boot, Spring Data JPA and PostgreSQL for two applications on a platform used by 10,000+ corporate clients.
- **You have real numbers:** card processing time cut **60%**, an audit system writing **50,000+ transactions a day, 1,000+ monthly reports**.
- **You have unusual depth for your level** — Camunda BPMN, AOP with `@Aspect`, Liquibase migrations, JasperReports, FileNet, Kafka and RabbitMQ. Most candidates at two years have CRUD APIs and nothing else.
- **You worked a real release process** — JUnit, SIT/UAT, Bamboo CI/CD, code review.

The one thing to manage: being full-stack can read as unfocused in a backend interview. Lead with backend, mention React once as useful context, and move on.

| • "Tell me about yourself" — your script

Now → What you built → Why this role. Under two minutes.

"I am a backend engineer with about a year and a half of production experience, most recently at Zensar on Standard Bank South Africa's Service Online platform, which is used by around 10,000 corporate clients.

I built the Spring Boot and PostgreSQL backends for two self-service applications — a business card application and an online transaction recall workflow. Both had multi-stage approval processes, so I modelled those as Camunda BPMN workflows, which took card processing time down by about 60%.

The piece I am most pleased with is an audit logging system I built with Spring AOP. The requirement was to capture every report download and email for compliance, and rather than adding a logging call to dozens of endpoints, I used `@Aspect` pointcuts so the audit logic lives in one class and the business code is untouched. It writes over 50,000 transactions a day to PostgreSQL.

I also worked on the React side of the same platform, so I have seen the API from the consumer's end — but backend is where I want to specialise, which is why this role interested me."

| Your four stories

Story 1 ●● — The AOP audit system (*lead with this*)

This is your best story because it is an **architecture** decision, not just a feature.

"The requirement was to audit every report download and email for compliance. The obvious approach is to add a logging call inside each endpoint, but that means touching dozens of files, and it is easy to miss one — and worse, easy for the next person adding an endpoint to forget.

So I used aspect-oriented programming. A pointcut matches the methods, and the aspect runs around them. The audit logic lives in one class, the business code is untouched, and it cannot be forgotten when someone adds a new endpoint. It writes more than 50,000 transactions a day to PostgreSQL."

Follow-ups to be ready for:

- "What is a pointcut versus an advice?" → A **pointcut** is the expression that selects which methods to intercept. An **advice** is the code that runs — `@Before` , `@After` , `@Around` . `@Around` is the one that can see both the arguments and the return value.
- "How does Spring AOP work underneath?" → **Proxies**. Spring wraps the bean in a proxy — a JDK dynamic proxy if it implements an interface, CGLIB otherwise — and the proxy runs the advice before delegating. **And the same caveat as `@Transactional` : an internal `this.method()` call bypasses the proxy, so the aspect does not fire.** Saying that unprompted is a genuinely strong moment.
- "What was the performance impact of writing 50,000 rows a day?" → Be honest about what you did. If the writes were asynchronous or batched, say so. If not, say what you would do now — an async write or a queue, so auditing never slows the request it is auditing.

Story 2 ● — The Camunda approval workflow

"The card application had a multi-stage approval flow. Rather than encoding the state machine in application code with status columns and conditionals scattered around, we modelled it as a BPMN process in Camunda. The engine tracks which step each application is at, so the approval rules live in one place and changing the process does not mean rewriting business logic. That took processing time down about 60%."

If they do not know Camunda, explain it simply: "It is a workflow engine — you define the process as a diagram and it manages the state transitions and the tasks waiting on people."

The general point to land, which matters more than the tool: *"the value is that the process is explicit and in one place, instead of being implied by status flags spread across the codebase."*

Story 3 — REST APIs and the frontend contract

You built REST APIs consumed by a 14-screen React frontend, and you also wrote that frontend. That is a useful angle few candidates have:

"Because I worked on both sides, I paid attention to the API as a contract. Consistent error shapes so the frontend has one thing to handle, proper status codes, and pagination on list endpoints. When you have been the consumer of a badly shaped API you design differently."

Story 4 — Testing and release

Unit unit and integration tests for REST APIs and workflow logic, validated through SIT and UAT. **The detail worth adding:** what you tested. Workflow state transitions are the valuable tests — that an approved application cannot be approved twice, that a rejected one cannot move back. Those are the rules that cost money.

●● The three questions your resume invites

1. ●● The gap — you left in February

Your resume shows **Dec 2024 - Feb 2026**, and it is now late September. **That is about seven months, and it is the first thing an interviewer will notice.** Prepare this properly; it matters more than any technical answer in this file.

Two or three sentences, factual, calm, no apology:

"[The real reason — the project rolled off, the client engagement ended, a personal reason.] Since then I have been [what you have actually done — deepening backend skills, building X, interviewing selectively]. I was clear that I wanted a backend-focused role rather than taking the next thing available, and this one matches that."

Fill in the truth. Do not invent anything. Project roll-offs and bench situations are extremely common in the services industry and no interviewer will hold one against you.

Then show the time was used. This is the part that turns the gap from a negative into a neutral or better:

- Be **specific**: "I went through the Spring Boot documentation properly and rebuilt X" beats "I was upskilling" by a mile.
- Your **GenAI Level 1 and 2 certifications** are dated 2025, so mention any learning since.
- If you have built anything, have it on GitHub and mention it.

⚠ **Do not let this be vague.** A seven-month gap with a clear story is fine. A seven-month gap with a mumbled answer is the thing that loses the interview.

2. • "You look like a frontend developer. Why backend?"

Your GitHub and portfolio are React-heavy, and your resume lists React under skills. Expect this.

"I have done both, and on the Service Online platform I did both on the same product. What I found is that the problems I enjoyed most were on the backend — the Camunda workflow modelling, the AOP audit design, getting the data model and transactions right. The frontend work I am glad to have done, because it made me better at designing APIs, but backend is where I want the depth."

That is credible because it explains the frontend rather than hiding it.

3. • The three things the JD asks for that your resume does not show

Prepare a one-line honest answer for each. None is disqualifying; being caught without an answer is what costs you.

Mockito. The JD names *"JUnit, Mockito, and unit testing frameworks"*. Your resume lists JUnit only.

"I have written JUnit unit and integration tests for REST APIs and workflow logic, validated through SIT and UAT. My mocking has been lighter than my JUnit usage, so I would not overstate it — but I understand the model." *(Then demonstrate it — file 15, Q3.)*

Cloud platforms. The JD mentions integrating with cloud platforms. You have Docker, not a cloud platform.

"Docker in the development setup and Bamboo for deployment, but I have not owned cloud infrastructure — exposure rather than depth. I understand the model: containerise the application and take configuration from the environment, so the same artefact is promoted from SIT to UAT to production."

Java 11 and 17. The JD says "Java 8/11/17+". You list Java 8.

"My production work has been Java 8. I have kept up with the LTS releases — records, text blocks, switch expressions and pattern matching in 17, `var` and the String utilities in 11." *(File 02, Q9. And connect records to Lombok, which you did use.)*

The pattern in all three: name the boundary, then show the understanding. That is a strong answer at 1–2 years. Bluffing and then being asked a second question is not.

4. The AI tools on your resume

You list Claude, GitHub Copilot and "LLM-assisted coding" under skills, plus GenAI certifications. At a services company that is more likely to be a **positive** — Infosys is pushing AI-assisted delivery hard — but be ready to answer well:

"I use them for a first draft, for explaining unfamiliar code, and as a second pair of eyes in review. But I do not commit anything I could not explain and defend myself, because I am accountable for it either way. And I would follow whatever the client's policy is on what can be pasted into an external tool — on a banking project that is a real constraint."

That second sentence is the one that matters. Say it.

✓ Before Saturday

1. Say the "tell me about yourself" script out loud, twice. Time it.
2. Write down the **real one-sentence reason** for the gap, and what you have done since. Say it until it sounds calm rather than defensive.
3. Rehearse the AOP story **including** the proxy follow-up.
4. Be able to say the numbers without hesitating: **60%, 50,000+ a day, 1,000+ monthly reports, 10,000+ clients**.
5. Have the "why backend" answer ready.

13 · HR, Managerial and the Offer Conversation

Time needed: 60 minutes. Say every answer out loud.

Infosys usually runs a separate managerial or HR round, and it is not a formality. People clear the technical round and lose the offer here.

1. "Why Infosys?"

Do not say "it is a big company" or "for growth". Say something only someone who read the posting could say.

"Three reasons.

The role is Java, Spring Boot and microservices, which is exactly what I have been doing and where I want to go deeper. That is a straightforward fit.

Second, I have banking domain experience from working on a UK private bank's systems, and Infosys works across financial services. I would rather build on a domain I already understand than start from nothing.

Third, the job description talks a lot about working with clients — requirement analysis, seeking clarifications, proposing alternatives. I have done that at Zensar with a client team in another country, and before that I spent a year in technical recruitment, so I am comfortable in those conversations. It is the part of the work I actually enjoy, and it seems central to how Infosys operates."

That third point is your real differentiator. Use it.

2. "Why are you leaving Zensar?"

Never criticise your current employer. Frame it as moving toward something.

"I have had good exposure at Zensar — I have owned features end to end on two banking platforms and worked across the full SDLC. What I am looking for now is more depth in microservices architecture rather than mostly working within one application, and more direct client-facing work. This role is explicitly microservices and consulting oriented, which is why I applied."

If asked about compensation, it is acceptable to say it is *a* factor, but never the first one you mention.

3. The JD's own language — prepare these four

The posting is full of consulting phrases. Each one is a question waiting to be asked.

"How do you handle unclear or incomplete requirements?" *(the JD's "problem definition" and "seek clarifications")*

"I do not start building on an assumption. I write down what I have understood, including the gaps, and take it back with specific questions rather than 'please clarify'. On the recall workflow there were states where it was not defined who could act, so I listed the cases, proposed a default for each, and asked them to confirm. That is much faster to answer than an open question, and it gets a decision recorded."

"How do you estimate effort?" *(the JD's "effort estimation")*

"I break the work into pieces small enough to reason about — API changes, data model changes, testing, integration — and estimate those rather than the feature as a whole. I add time explicitly for testing and for integration issues, because those are what people forget. And I flag what the estimate assumes, so if an assumption turns out to be wrong the estimate can be revisited rather than just being missed."

"How do you learn a new technology?" *(the JD's "research, literature surveys, POCs")*

"Official documentation first, because tutorials teach you one path and the docs tell you the model. Then I build a small proof of concept, because reading does not surface the problems. When I picked up Kafka I built a small producer and consumer before touching the real system, which is where I actually understood consumer groups and offsets. Then I check how it is used in the existing codebase, since conventions there matter more than my preferences."

"How do you approach diagnosing a production issue?" *(the JD's "diagnose the root cause")*

Use your real process from file 10, Design 5: reproduce if possible, check logs and timings, form a hypothesis, verify it before fixing, fix, then add something that would have caught it earlier. Finish with a concrete example.

4. Standard behavioural questions — use STAR

Situation → **Task** → **Action** → **Result**. Under 90 seconds. Always end on a result.

"Tell me about a challenging technical problem." → The AOP audit system, or the Camunda approval workflow. Both are in file 11.

"A time you disagreed with someone." → Best if it is a technical disagreement you resolved with evidence rather than opinion. The pattern that works: you raised the concern, then brought data or a working alternative instead of just objecting.

"A time you made a mistake." → Have one. Refusing to answer this looks worse than the mistake. Pick something real, own it plainly, and finish with what you changed. A missed edge case caught in QA, with "and I added a test for that state transition afterwards", is a good shape.

"**How do you handle pressure or a tight deadline?**" → Scope, communicate early, do not silently slip. You worked in SIT and UAT cycles, so you know what happens when something half-finished reaches testing.

"**What is your weakness?**" → Real, bounded, with the fix.

"Coming from a non-linear route into development, I was initially slower at reading large unfamiliar codebases than people who had been doing it longer. I have got much better by tracing one request end to end through the code before changing anything, rather than jumping straight to the file I think is responsible."

Never "I am a perfectionist" or "I work too hard".

"**Where do you see yourself in 3 years?**"

"Owning a service or a module end to end, and being the person on the team who is trusted with performance and data-layer problems. Longer term I would like to move toward solution design, which is part of why the consulting side of this role appeals."

"**How do you work in a team?**" → Mention code reviews (you do them), QA coordination (on your resume), and being specific in reviews rather than vague.

5. Logistics questions — answer instantly

Question	How to answer
Notice period	State it exactly. If it is 60 or 90 days, say so, and mention whether buyout is possible. Do not guess.
Willing to relocate?	Your resume already says "open to relocate anywhere in India / remote", so say it clearly and early — it is a real advantage with Infosys, which staffs across Hyderabad, Bangalore, Pune, Chennai and Mysore. Just make sure it is actually true, because they will hold you to it.
Work from office?	Infosys is largely back in office. Answer directly.
Shifts / on-call?	Client projects sometimes need overlap with UK or US hours. You already work with a UK client, so say that — you know what it involves.
Do you have other offers?	Honest and brief. Do not invent leverage; it usually backfires.

⚠ If any of these are genuinely a problem, say so **now**, politely. Accepting and renegotiating later damages the relationship and your reference.

6. Salary

Know your numbers before the call: your current fixed CTC, your current total CTC, and the number you want. Do not do arithmetic live.

When asked for expectations:

"My current CTC is [X]. Based on my experience with Java, Spring Boot, microservices and messaging, and my banking domain background, I am looking for [Y]. But I am interested in the role and the project, so if the fit is right I am open to discussing it."

Rules:

- **Never lie about your current CTC.** Infosys verifies with payslips and Form 16. Being caught costs the offer, usually after you have resigned elsewhere.
- **Ask for the breakup** at offer stage: fixed, variable, retention bonus, and what the in-hand figure is. Large companies often put a meaningful share in variable pay.
- **Ask about the band or level.** Infosys has defined levels, and the band affects your future increments more than the starting number does.
- If they ask first, it is completely normal to ask **"what range have you budgeted for this role and level?"**

7. • Questions YOU ask them

Have four ready, ask two or three. Ending with "no, I'm good" reads as no interest.

Technical:

1. "Which domain would this role sit in — is it financial services, given the skills?"
2. "Is the project greenfield microservices, or modernising an existing monolith?"
3. "What does the current stack look like — Spring Boot version, cloud platform, CI/CD?"
4. "How large is the team, and would I own a service end to end?"

Role and growth:

5. "What would a successful first three months look like?"
6. "Is there scope to move toward design and client-facing work over time?"
7. "How does Infosys handle certification and upskilling for someone at this level?"

The one that makes you memorable, for the managerial round:

"What is the biggest technical problem the team is dealing with right now that this hire would help with?"

Then **listen**, and connect your answer to it. That is how you close.

8. On the day

- Two printed copies of your resume, and a copy open on your laptop.
- **Know your own resume cold.** Everything on it is fair game — Kafka, RabbitMQ, Eureka, JWT, Hibernate lazy loading, MongoDB, execution plans. If you touched something briefly, say so honestly rather than overstating.
- Carry your ID, and have your employment and education documents ready — Infosys onboarding is document heavy.

- If it is online: join five minutes early, test camera and microphone, sit somewhere quiet with a plain background, and keep an offline copy of your notes in case the network drops.
- Dress smart. Slightly overdressed beats underdressed.

| 9. If you get stuck

- **Do not freeze — narrate.** "Let me think about that for a second" is a complete and acceptable sentence.
- **Do not bluff.** "I have not used that directly. My understanding is X — is that the direction you mean?" has never cost anyone an offer. Bluffing has, because the next question exposes it.
- **Ask what they are getting at.** "Do you mean at the database level or in the application?"
- If you fumble one question, let it go. People lose interviews by replaying question 4 while answering question 7.

14 · Delivery, Code Review and Working With Clients

Time needed: 60 minutes

| First, the context: read this job description again

Almost the entire responsibilities section of this posting is about things that are **not writing code**:

"interface with the client for quality assurance, issue resolution and ensuring high customer satisfaction... understand requirements, create and review designs, validate the architecture... participate in project estimation, provide inputs for solution delivery, conduct technical risk planning, perform code reviews and unit test plan reviews... lead and guide your teams towards developing optimized high quality code deliverables, continual knowledge management and adherence to the organizational guidelines and processes."

The only technical requirement listed is **Java and Spring Boot**, as both primary and preferred skill. Everything else in that block is about **how you deliver**.

What that tells you: this is not a pure coding role. It is Infosys's analyst-shaped template — someone who can be put in front of a client, review other people's work, and be trusted to follow process. The technical round will still test Java and Spring Boot properly, but there will be a second conversation about all of the above, and most candidates walk into it with nothing prepared.

The good news: your resume already contains evidence for nearly every item. Code reviews, QA coordination, requirement analysis, deployment validation, production incident resolution. You have done these things. What you need is to have the answers **ready**, and to describe them as ownership rather than participation.

⚠ **One framing note.** You have under two years of experience, so do not claim to have led a team. Claim what is true: you reviewed code, you coordinated with QA, you handled production issues. Then show you understand what the next level up involves. Interviewers respect that far more than an inflated claim they can puncture with one follow-up.

| Q1. • "Have you done code reviews? What do you look for?"

Near-certain, because the JD names it explicitly.

Say this.

"Yes, code reviews were part of our normal process — both getting mine reviewed and reviewing others'.

I look in a rough order of importance. First **correctness**: does it do what the requirement says, and are the edge cases handled — nulls, empty results, boundary values. Second **error handling**: are exceptions handled at the right level, or is something being swallowed silently, which is the worst outcome because the failure becomes invisible.

Then **the data layer**, because that is where I have found the most real problems. Is there an N+1 sitting in a loop, is a query missing an index it will need, is `@Transactional` on the right method. Then **tests** — not the count, but whether the meaningful cases are covered. And last **readability**: naming, method length, duplication. That matters, but I would not block a change on it alone."

The follow-up: "How do you give the feedback?" This is really a communication question.

"I separate what must change from what is a suggestion, and say which is which — a comment that just says 'this could be better' wastes everyone's time. I ask rather than instruct when I am not sure of the context, because often there is a reason I cannot see. And if a thread goes past two or three replies I stop typing and talk to the person, because that usually means we are misunderstanding each other rather than disagreeing."

Real-time example. In a card management change, a reviewer's comment on my code was that a status update was not inside the transaction boundary, so a failure partway through would have left the record updated but the audit row missing. That was a correctness bug a test would not have caught, and it is the kind of thing I look for now when I review someone else's.

Q2. • "What is a unit test plan review? What makes a test plan good?"

Explicitly in the JD, and most candidates have never thought about it.

Say this.

"A test plan review is checking that the tests cover the right things **before** the code is written or signed off, rather than discovering the gap in UAT.

What I look for is whether the tests cover **behaviour and rules**, not just lines. A suite with high coverage that only tests the happy path is worse than a smaller suite that tests the rules that actually matter. So I check three things: is the happy path covered, are the **failure and boundary cases** covered, and are the **state transitions** covered — the transitions that must be **prevented** as much as the ones that are allowed."

Real-time example — this is directly on your resume.

"On the business card workflow I wrote unit tests for state transition validation. The important tests were not 'an approval succeeds' — they were the negative ones: an already approved request cannot be approved again, a rejected request cannot move back to pending, and a requester cannot approve their own request. Those are the rules that cost money if they break, and they are exactly what a test plan review should be checking for."

If they ask about coverage targets:

"Coverage tells you what was executed, not what was verified. A test with no assertion still counts towards it. I would rather aim for the critical paths being genuinely tested than chase a percentage, though a very low number is still a useful warning sign."

Q3. • "How do you estimate effort?"

In the JD as "participate in project estimation". Give a method, not a number.

Say this.

"I break the work down until the pieces are small enough that I can actually picture doing them — API changes, data model changes, the tests, and the integration. Estimating a whole feature in one number is guessing; estimating six small pieces is much closer.

I explicitly include the things people leave out: testing, code review turnaround, and integration issues. Integration is where estimates fail most often, because it is the part that depends on other people.

And I state the assumptions with the estimate. If I have assumed the API contract is fixed and it then changes, the estimate should be revisited rather than just missed — stating the assumption is what makes that conversation possible instead of it looking like a slip."

Real-time example. A change that looks like one field on a screen is rarely one field: it is the DTO, the validation, the entity and migration, the query, the tests, and the downstream event if anything consumes it. Walking that list is how I avoid estimating the visible part only.

If they ask about techniques: mention that you have worked in Agile with story points and relative sizing, and that the value of a group estimate is mostly in surfacing the different assumptions people are holding, not in the number itself.

Q4. • "What is technical risk planning? Give me an example."

The JD phrase is "conduct technical risk planning". Most people freeze on this. It is simpler than it sounds: **what could go wrong, how likely, how bad, and what we do about it.**

Say this.

"It is identifying what could go wrong technically before it does, and deciding on the response in advance rather than during the incident.

On a change I would ask: what does this depend on that is outside my control; what happens under a load much higher than we test with; what is the blast radius if it fails; and how would we roll it back. Then for the serious ones, either mitigate it in the design or agree a fallback."

Real-time example — use your own migration.

"When we moved email notifications from synchronous sending to RabbitMQ, the obvious risk was that the queue or consumer goes down and notifications silently stop — which is worse than the timeout we were fixing, because a timeout is visible and silence is not. So the mitigations were retries with backoff, a dead letter queue so a failing message moves aside instead of blocking the queue, and alerting on the DLQ so somebody actually finds out. Asynchronous processing without a DLQ and monitoring just moves the failure somewhere nobody is looking."

That answer is strong because the risk is real, and the mitigations are specific.

| Q5. • "How do you review a design, or validate an architecture?"

The JD says "create and review designs, validate the architecture". You are not expected to be an architect. You are expected to ask sensible questions.

Say this.

"I check it against the requirement first — does the design actually do what was asked, and are the non-functional parts stated: expected volume, response time, what has to be consistent versus what can be eventual.

Then the practical questions. Where does the data live and who owns it. What happens when a dependency is unavailable. Is anything a single point of failure. How is it secured, and is authorisation enforced server side. How will we know it is working in production — logging, correlation IDs, metrics. And how does it get released and rolled back.

The last one I ask is whether it is more complex than the problem requires. Simplicity is a real design property, and complexity that is not paying for itself is a cost forever."

Real-time example. Publishing transaction state changes to Kafka rather than calling the compliance and reporting services directly was a design decision with a trade-off: it removed the coupling and let each consumer read at its own pace, at the cost of eventual consistency and a harder debugging story. Being able to state the cost of your own design, not just its benefit, is what a design review is for.

Q6. • "How do you handle a client raising a quality issue?"

The JD's first line is client interfacing for quality assurance and issue resolution. You worked with a UK bank, so you have real material.

Say this.

"First, acknowledge quickly and separate the two things: what we do to contain it now, and what caused it. Clients get much more anxious about silence than about the problem itself.

Then get to the actual facts before promising anything — reproduce it, check the logs and what changed recently. I do not commit to a cause or a timeline until I know, but I do commit to when I will come back with an update, and I keep that.

Then fix, verify, and say plainly what happened and what we have changed so it does not recur. That last part is what actually restores confidence."

Real-time example. Production incidents I worked on were diagnosed by analysing SQL execution plans and application logs to find the root cause, then coordinating the fix with QA before it went out. The important part was not the fix, it was that QA validated it in SIT before it reached the client again.

The line that lands with a services company:

"Being wrong quickly and openly costs far less than being confident and wrong slowly."

Q7. "How do you guide or mentor someone more junior?"

Answer honestly at your level.

Say this.

"I have not formally led a team, so I will answer about what I have actually done. Most of it has been through code review and pairing on problems. What I have learned is that pointing at the answer does not stick — explaining why, or asking the question that gets them there, does.

The other thing is writing things down. If I explain the same thing twice, that is a documentation gap, not a person problem."

That answer is credible, and the honesty at the start makes the rest believable.

Q8. "Knowledge management" and "adherence to processes" — what do these mean?

Both are literally in the JD, and both sound like filler until you answer them concretely.

Knowledge management — say this.

"Making sure what I know is not only in my head. In practice that is API documentation that is actually current — we used Swagger, so it lives with the code rather than drifting in a separate document — plus runbooks for things like what to check when the queue backs up, and proper handover notes when someone rotates off. The test is whether someone else could pick up my work if I were away, and I would rather find that out deliberately than during an emergency."

Adherence to processes — say this.

"In a bank the process exists for a reason, and the reason is usually that something went wrong once. We had a defined SDLC — requirement analysis, design, implementation, unit tests, then SIT and UAT before release — with code review and deployment validation as gates. I have never found the process to be the bottleneck; what slows things down is ambiguity in requirements, which is why I push to get those pinned down early."

Q9. "How do you handle unclear or changing requirements?"

Say this.

"I do not start building on an assumption I have not confirmed. I write down what I have understood, including what is missing, and go back with **specific** questions rather than asking them to clarify — a specific question with a proposed default is much faster for someone to answer, and it gets a decision on record.

When requirements change mid-work, the important thing is to say immediately what it means for the estimate and what else it affects, rather than quietly absorbing it. Absorbing changes silently is how a project ends up late with nobody able to explain why."

Real-time example. On the recall workflow there were states where it was not defined who was allowed to act. Rather than pick something, I listed the cases, proposed a default for each, and asked for confirmation. That took a short call instead of a long email thread, and it meant the rules were written down rather than living in the code.

Q10. "What makes code high quality?" (the JD says "optimized high quality code deliverables")

Say this.

"Four things, roughly in order.

It is **correct**, including the cases that are not the happy path. It is **readable**, because it will be read far more often than it is written, and the next person may be me in six months. It is **tested** where it matters, particularly the rules that cost money if they break. And it **performs acceptably under real data volumes**, not just the ten rows in a developer's local database — which is the specific mistake behind both of the performance problems I fixed, because both were fine in testing and slow in production.

The one I would add for a bank is **traceability**. If something goes wrong, can we tell what happened, who did it and when. That is why proper logging and an audit trail are part of the deliverable, not an extra."

✓ Check yourself before moving on

1. What you look for in a code review, **in order**, and how you give the feedback.
2. What makes a good unit test plan — and the negative state-transition example.
3. Your estimation method, including stating assumptions.
4. The RabbitMQ risk-planning example, with the DLQ and alerting.
5. How you handle a client-raised quality issue, in four steps.

15 · JUnit, Mockito, Build Tools and Agile

Time needed: 60 minutes

The job description names all of this explicitly, and it is the part of the spec your resume covers most thinly. Your resume lists JUnit but **not Mockito**, and the JD asks for *"JUnit, Mockito, and unit testing frameworks"*. Close that gap before Saturday — it is the cheapest marks available this week.

The JD also names **Maven or Gradle, Git, Agile ceremonies, code reviews, technical documentation** and **supporting production deployments**. None of those is hard. All of them are questions you can lose points on by having nothing to say.

Part A · JUnit and Mockito ●●

| Q1. Unit test vs integration test

Say this.

"A unit test exercises one class in isolation with its collaborators mocked, so it is fast and when it fails you know exactly what broke. An integration test wires real things together — the repository against a database, or the controller through the Spring context — and tells you the pieces actually fit.

I write mostly unit tests over service logic, because that is where the business rules live, plus a smaller number of integration tests over the paths that would be expensive to get wrong. What I test is **behaviour and rules**, not lines."

Real-time example from your own work. On the card approval workflow, the valuable tests are not "an approval succeeds". They are the **negative state transitions**: an already approved application cannot be approved again, a rejected one cannot move back to pending, a requester cannot approve their own. Those are the rules that cost money, and they are exactly what a workflow needs tested.

Q2. ●● JUnit 5 — the annotations

```
class CardApplicationServiceTest {

    @BeforeEach void setUp() { } // before EVERY test
    @AfterEach void tearDown() { }
    @BeforeAll static void once() { } // once for the class - must be static
    @AfterAll static void done() { }

    @Test
    void approvesApplicationWhenAuthoriserIsValid() { }

    @Test @Disabled("pending requirement clarification")
    void skipped() { }

    @ParameterizedTest
    @ValueSource(strings = {"", " ", " "})
    void rejectsBlankReference(String input) { }
}
```

JUnit 4 → 5 differences, in case they ask: `@Before` became `@BeforeEach`, `@BeforeClass` became `@BeforeAll`, `@RunWith` became `@ExtendWith`, `@Ignore` became `@Disabled`, and expected exceptions moved from `@Test(expected = ...)` to `assertThrows`.

```
assertEquals(APPROVED, result.getStatus());
assertTrue(result.isComplete());
assertThrows(IllegalStateException.class, () -> service.approve(alreadyApproved));
assertAll(() -> assertEquals(1, a), () -> assertEquals(2, b)); // reports both failures
```

Q3. ●● Mockito — the core of this file

```
@ExtendWith(MockitoExtension.class)
class CardApplicationServiceTest {

    @Mock CardApplicationRepository repository; // database - mock it
    @Mock NotificationSender notifier; // external - mock it
    @InjectMocks CardApplicationService service; // the class under test

    @Test
    void notifiesRequesterOnApproval() {
        CardApplication app = new CardApplication(1L, PENDING);
        when(repository.findById(1L)).thenReturn(Optional.of(app));

        service.approve(1L, "authoriser-7");

        ArgumentCaptor<CardApplication> saved = ArgumentCaptor.forClass(CardApplication.class);
        verify(repository).save(saved.capture());
        assertEquals(APPROVED, saved.getValue().getStatus());
        verify(notifier).send(eq("authoriser-7"), any(Message.class));
    }

    @Test
    void throwsWhenApplicationMissing() {
        when(repository.findById(99L)).thenReturn(Optional.empty());
        assertThrows(ApplicationNotFoundException.class, () -> service.approve(99L, "x"));
        verifyNoInteractions(notifier); // and nothing was sent
    }
}
```

The pieces, and what each is for:

What it does	
@Mock	Creates a fake collaborator
@InjectMocks	Builds the class under test with the mocks injected
@Spy	Wraps a real object — most calls real, some stubbed
@MockBean	The Spring version: replaces a bean in the application context
when(x).thenReturn(y)	Stub the input
when(x).thenThrow(e)	Stub a failure
verify(mock).method()	Assert the interaction happened
verify(mock, times(2))	...this many times. Also <code>never()</code> , <code>atLeastOnce()</code>
verifyNoInteractions(mock)	Assert nothing was called on it

What it does	
<code>ArgumentCaptor</code>	Capture what was passed, to assert on it
<code>any()</code> , <code>eq()</code>	Argument matchers — if you use one, use them for all arguments

Say this.

"I mock what I do not own or what would make the test slow and non-deterministic — repositories, external clients, anything over the network. I do not mock the class under test, and I try not to mock simple value objects, because then the test just asserts my own mock setup back at me.

`when(...).thenReturn(...)` sets up the input and `verify(...)` checks an interaction actually happened, which matters when the important outcome is a side effect rather than a return value. `ArgumentCaptor` is what I reach for when I need to assert on what was passed, not just that something was."

● The questions they follow up with

"Mock vs stub?"

"A stub just returns canned answers so the test can proceed. A mock also records how it was called, so you can verify the interaction. Mockito does both, which is why the words get used loosely."

"What can Mockito not mock?" Historically `static` and `final` methods and private methods — `mockito-inline` handles statics now, but the better answer is that needing to mock a static is usually a design smell, because it means the dependency is not injectable.

"How do you test a void method?" You cannot assert a return value, so you verify the interaction — `verify(repository).delete(entity)` — or capture and assert on the argument.

"How do you stub a void method to throw?" `doThrow(ex).when(mock).method();` — the `doX().when()` form, because `when(mock.voidMethod())` will not compile.

Q4. Testing a Spring Boot application

```
@WebMvcTest(CardController.class)           // controller layer only - fast
class CardControllerTest {
    @Autowired MockMvc mockMvc;
    @MockBean CardApplicationService service;

    @Test void returns404WhenMissing() throws Exception {
        when(service.findById(1L)).thenReturn(new ApplicationNotFoundException(1L));
        mockMvc.perform(get("/api/cards/1")).andExpect(status().isNotFound());
    }
}
```

- `@SpringBootTest` loads the whole context — integration, slow.
- `@WebMvcTest` / `@DataJpaTest` load one slice — much faster.
- `@MockBean` swaps a real bean for a mock inside the context.

Say why slices matter: "A fast suite is one people actually run. If every test boots the whole application, the suite gets skipped, and a skipped suite protects nothing."

Q5. Coverage — the answer that shows judgement

"Coverage tells you what was **executed**, not what was **verified** — a test with no assertion still counts towards it. So I would rather have the critical paths genuinely tested than chase a percentage. That said, a very low number is a useful warning sign, and a quality gate on new code in the pipeline is a reasonable safety net."

Q6. • If they ask how much Mockito you have used — be honest

Your resume says JUnit, not Mockito. If the question is direct:

"I have written JUnit unit and integration tests for REST APIs and workflow logic, and validated them through SIT and UAT. My mocking has been lighter than my JUnit usage, so I would not overstate it — but I understand the model: mock the collaborators you do not own, stub with `when/thenReturn`, verify interactions, and capture arguments when you need to assert on what was passed."

Then show it using the example above. That answer is completely respectable at your level. Claiming deep Mockito and then fumbling `doThrow` on a void method is not.

Part B · Build tools and Git

Q7. Maven — what they will actually ask

"Maven is a build and dependency management tool driven by `pom.xml`. It resolves dependencies transitively from a repository, and it defines a standard lifecycle so any Maven project builds the same way."

The lifecycle, in order — worth knowing:

`validate` → `compile` → `test` → `package` → `verify` → `install` → `deploy`

- `mvn clean install` — the one everybody runs. Wipes `target/`, builds, runs tests, and puts the artefact in your **local** repository.
- `mvn package` — builds the JAR but does not install it locally.
- `mvn test` — compiles and runs tests only.

Scopes: `compile` (default), `provided` (available at compile, supplied at runtime — e.g. a servlet API), `runtime`, `test` (JUnit, Mockito).

Two things worth mentioning unprompted:

- **The parent POM.** `spring-boot-starter-parent` manages versions, which is why you add `spring-boot-starter-web` with no version and get a compatible set.
- **Dependency conflicts.** Maven resolves to the **nearest** version in the tree, which is how you end up with a version you did not ask for. `mvn dependency:tree` is how you find it; `<dependencyManagement>` or an exclusion is how you pin it. Saying this signals real build experience.

Maven vs Gradle: Maven is XML and convention-driven, predictable and verbose. Gradle uses a Groovy or Kotlin DSL, is more flexible, and is faster on large builds thanks to incremental builds and a build cache. You use Maven; say so, and say you could pick up Gradle.

Q8. Git — the likely questions

- **merge vs rebase** — merge preserves history and adds a merge commit; rebase replays your commits on top for a linear history. **Never rebase a shared branch**, because it rewrites commits other people already have.
- `git pull` = `fetch` + `merge`. `fetch` alone just updates your remote-tracking branches, which is the safe way to look before you leap.
- **Resolving a conflict** — understand both sides rather than picking one. In business logic, taking one blindly silently drops someone's work.
- **revert vs reset** — `revert` makes a new commit undoing an old one, safe on a shared branch. `reset` moves the branch pointer and rewrites history, so it is for local work only.
- **cherry-pick** — apply one commit onto another branch, typically a hotfix.
- **Branching** — feature branch, pull request, reviewed before merge. You used Bitbucket and Bamboo, so say that.

Part C · Agile, documentation and deployment

The JD lists these explicitly, so have a sentence for each.

Q9. The ceremonies — what each is actually for

Ceremony	Purpose	The honest thing to say
Sprint planning	Agree what the team commits to	"This is where ambiguity should be caught — if a story is not clear enough to estimate, it is not ready"
Daily stand-up	Surface blockers, not a status report	"The useful part is the blockers. If it turns into a status recital it stops being worth the time"

Ceremony	Purpose	The honest thing to say
Sprint review	Show working software to stakeholders	"Demoing to the BA and product owner is where you find out you built the wrong thing — better than in UAT"
Retrospective	Improve the process	"The value is only in the actions. A retro with no action items is a complaint session"
Refinement	Size and clarify upcoming work	"Where most estimation problems get solved, before anything is committed"

Say this if asked generally:

"We worked two-week sprints with stand-ups, refinement, review and retro, tracked in Jira. The ceremony I found most valuable was refinement, because that is where an unclear requirement gets caught — before someone has spent three days building the wrong thing."

Q10. "Analyse business requirements and translate them into technical solutions"

Straight from the JD's responsibilities. It is really a question about handling ambiguity.

"I do not start building on an assumption I have not confirmed. I write down what I have understood including the gaps, then go back with **specific** questions and a proposed default for each, rather than asking someone to clarify. A specific question is much faster to answer and it gets the decision recorded.

On the recall workflow there were states where it was not defined who could act. I listed the cases, proposed a default for each, and asked the BA to confirm. That was a short call instead of a long email thread."

Q11. Technical documentation

"API documentation that lives with the code, so it does not drift — we used Swagger, so the contract is generated from the code rather than maintained separately. Beyond that, a short design note for anything non-obvious, and runbook-style notes for operational things. My test is whether someone else could pick up my work if I were not there."

Q12. Supporting production deployments

"We released through Bamboo CI/CD after SIT and UAT sign-off. What I would do around a release: validate the deployment in the environment rather than assuming it worked, watch the logs and error rates for the first period after it goes out, and know what the rollback is **before** you deploy, not while something is failing.

Database changes need particular care, which is why we used Liquibase — versioned migrations mean the schema change is repeatable and reviewable rather than someone running a script by hand."

That Liquibase point is a genuinely strong one and it is on your resume. Use it.

Q13. Cloud platforms — the JD mentions integration, be honest

Your resume has Docker but not a cloud platform.

"I have worked with Docker in the development setup, and our deployments went through Bamboo, but I have not owned cloud infrastructure — so I would call that exposure rather than depth. I understand the model: the application is containerised and configuration comes from the environment rather than the artefact, which is what lets the same build be promoted from SIT to UAT to production."

That last sentence shows you understand the principle even without the platform experience, which is what they are actually probing.

✓ Check yourself before moving on

1. Write the Mockito test from memory — `@Mock`, `@InjectMocks`, `when`, `verify`, `captor`.
2. `doThrow(...).when(mock).method()` — and **why** `when(mock.voidMethod())` will not compile.
3. The Maven lifecycle in order, and what `mvn clean install` does.
4. merge vs rebase, and why you never rebase a shared branch.
5. What each Agile ceremony is for, in one sentence each.
6. Your honest Mockito answer, and your honest cloud answer.

16 · Final Cheat Sheet — SATURDAY MORNING ONLY

45 minutes. Nothing else. No new material.

| The 10 answers that carry this interview

1. **HashMap:** "`hashCode()` , then a spread of `h ^ (h >>> 16)` , then bucket index `(n-1) & hash` . Collisions chain in a list; since Java 8 a bucket over 8 entries becomes a red-black tree. Capacity 16, load factor 0.75, so it resizes and rehashes at 12."
 2. **equals/hashCode:** "Equal objects must have equal hash codes. Override `equals` without `hashCode` and the map looks in the wrong bucket and never finds the entry."
 3. **N+1:** "One query for the list, then one per row for the relation. Find it by counting queries in the SQL log. Fix with JOIN FETCH, `@EntityGraph` , batch fetching or a DTO projection."
 4. **@Transactional internal call:** "It works through a proxy. An internal `this.method()` bypasses the proxy, so no transaction starts." Rolls back on **unchecked** exceptions only. **The same is true of your AOP aspect.**
 5. **Constructor injection:** "Fields can be final, dependencies cannot be missing, and I can test with `new Service(mock)` without a Spring context."
 6. **Spring AOP:** "A pointcut selects the methods, an advice is the code that runs. Spring wraps the bean in a proxy — JDK dynamic proxy with an interface, CGLIB otherwise."
 7. **Brute force first:** "Two nested loops is $O(n^2)$ — a HashMap makes the lookup $O(1)$, so one pass, $O(n)$. Trading space for time."
 8. **Binary search:** `lo + (hi - lo) / 2` to avoid overflow, `while (lo <= hi)` , only on sorted data, $O(\log n)$.
 9. **Microservices trade:** "You buy independent deployment, scaling and fault isolation; you pay with network failure, no distributed transaction and harder debugging."
 10. **Index cost:** "Faster reads, but every insert and update maintains every index, so I add them deliberately."
-

| DSA — the six patterns

- **Hashing** → counting, lookup, duplicates, two sum, anagram. *If you are searching inside a loop, a HashMap probably makes it one pass.*
- **Two pointers** → palindrome, reverse in place, pair-with-sum in a **sorted** array. $O(1)$ space.
- **Sliding window** → max sum of size k, longest substring without repeats. *Slide: add the new, drop the old.*
- **Fast and slow pointers** → middle of a list, cycle detection (Floyd's).

- **Stack** → balanced brackets, next greater element. Use `ArrayDeque`, **not** the legacy `Stack` class.
- **Binary search** → sorted data only.

Always: restate → ask edge cases → say the brute force and its cost → improve → write, talking → trace one example → **state time and space**.

Edge cases to ask about every time: null, empty, one element, duplicates, all-equal.

Complexity: `ArrayList` get $O(1)$, search $O(n)$ · `LinkedList` get $O(n)$ · `HashMap` $O(1)$ avg · `TreeMap` $O(\log n)$ · sorting $O(n \log n)$ · binary search $O(\log n)$.

Reverse a linked list — three pointers, save `next` *before* you flip:

```
next = curr.next; curr.next = prev; prev = curr; curr = next;
```

Java traps: `Integer` cache is $-128..127$, so `128 == 128` is **false** boxed ·

```
new String("a") != "a" · 0.1 + 0.2 != 0.3, money is BigDecimal.
```

Core Java

- **Overloading** compile time, different params · **overriding** runtime, subclass. Static methods are **hidden**, not overridden.
- **Abstract class** = shared state and code, one only · **interface** = contract, many.
- **HashMap** not thread safe · **Hashtable** locks the whole map, legacy · **ConcurrentHashMap** locks per bucket.
- **String immutable** — pool, security, thread safety, cached hash.
- **Checked** = compiler forces it · **unchecked** extends `RuntimeException`.
- `finally` always runs except `System.exit()`. Never `return` from it.
- **Fail-fast** throws `ConcurrentModificationException` — remove via `iterator.remove()`.

Java 8

- **Intermediate** lazy (`filter`, `map`) · **terminal** triggers (`collect`, `reduce`).
- `map` one-to-one · `flatMap` one-to-many then flattened.
- `groupingBy(X, counting())` · `groupingBy(X, summingDouble(Y))`.
- **Optional:** `map` / `orElse` / `orElseThrow`, not `isPresent` + `get`.
- ⚠ Group in **SQL** when the data is in the database, not in a stream over a million rows.

Spring Boot

- **IoC** = framework controls creation · **DI** = how it supplies dependencies.
- Beans are **singleton by default** → keep them **stateless**. A mutable field is a concurrency bug and, on a banking platform, an audit failure.
- `@Repository` also **translates database exceptions**.
- `@RestControllerAdvice` = one error shape. Log the detail, return a generic message.

- **Auto-configuration** = `@ConditionalOnClass` + `@ConditionalOnMissingBean` ; your bean wins.
- `@WebMvcTest` slice · `@SpringBootTest` full · `@MockBean` replaces a bean.

JPA and Hibernate

- **JPA** spec · **Hibernate** implementation · **Spring Data JPA** the layer on top.
- `@ManyToOne` / `@OneToOne` default **EAGER** — set them **LAZY**. `@OneToMany` / `@ManyToMany` **LAZY**.
- **Dirty checking** — a managed entity needs no `save()` .
- `@Enumerated(EnumType.STRING)` , never **ORDINAL**. Money is **BigDecimal**.
- **REQUIRES_NEW** = the audit row survives when the main transaction rolls back.
- **Liquibase** = versioned schema migrations, so the schema is reproducible per environment.

SQL

- **Left-most prefix**: an index on `(status, created_at)` does not serve `created_at` alone.
- Index killers: a **function on the column**, a **leading %** in LIKE.
- **EXPLAIN** : `type: ALL` = full scan. Check `key` , `rows` , `Extra` .
- **WHERE** before grouping · **HAVING** after. **UNION ALL** is faster than **UNION** .
- Customers with no orders = `LEFT JOIN ... WHERE t.id IS NULL` .

Your stories — the numbers

Standard Bank South Africa, Service Online · 10,000+ corporate clients.

AOP audit system — `@Aspect` pointcuts, **50,000+ transactions a day**, no change to business logic. *Your best story: it is an architecture decision, not a feature.*

Camunda BPMN approval workflow — card processing time down **60%**.

JasperReports — 1,000+ monthly reports. **FileNet** for documents. **Liquibase** migrations. **Kafka and RabbitMQ** messaging. **JUnit**, SIT/UAT, Bamboo CI/CD.

The AOP follow-up to have ready: pointcut selects, advice runs; Spring uses a **proxy**, so an internal `this.method()` call does not trigger the aspect — same as `@Transactional` .

The three questions you must not fumble

"Tell me about yourself" → Now → What you built → **Why this role**. Under 2 minutes.

"You left in February — what have you been doing?" → Two or three sentences, the **real** reason, then **specifically** what you have done since. Calm, no apology. A seven-month gap with a clear story is fine; a mumbled one is not. **Rehearse this more than any technical answer.**

"You look like a frontend developer. Why backend?" → *"I did both on the same product. The problems I enjoyed most were backend — the workflow modelling, the AOP design, the data*

model. The frontend work made me better at designing APIs, but backend is where I want the depth."

| Questions to ask them

"What would I be working on in the first few months?" · "What does the stack look like — Spring Boot version, cloud, CI/CD?" · "How does code review work in the team?" ·

"What is the biggest technical problem the team is dealing with right now?"

| The last five minutes

- **You are immediately available.** Most candidates serve 60–90 days. Say it.
- **Give the why, not just the what.** "HashMap stores key-value pairs" is textbook; "O(1) because it hashes to a bucket instead of scanning" is an engineer.
- **If you do not know something: say so, then reason out loud.** That has never cost anyone an offer. Bluffing has.
- In the coding round, **keep talking**. Silence reads as stuck.
- Sit up. Breathe out slowly. Smile before you speak.

Go get it.